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Developing and Implementing a Sexually Transmitted Disease Screening Program to Decrease Chlamydia Cases on a College Campus

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Abstract

Introduction: Sexually active college students were experiencing a high incidence, spread, and reinfection of the sexually transmitted Disease (STD) chlamydia on a university campus. **Purpose:** To determine if implementing of a quality improvement program on campus utilizing a holistic approach would help decrease chlamydia cases. This program focused on fast-track STD screening, increasing knowledge about STDs, and addressing mental health issues. **Methods & Study Design:** This was a descriptive/quasi-experimental design. Participants utilized the fast-track STD screening and completed surveys. Descriptive statistics and frequency tables were used to summarize survey responses for 1) the pre-and post-sexually transmitted disease (STD) education survey, and 2) the fast-track STD screening and sexual health questionnaire. Demographical data and screening results were also analyzed. The target participants were young adults inclusive of all genders between the ages of 18-25. The sample size for the fast -track STD screening and STD education was 31. **Results:** During the fast-track screening program there was 31 participants. Of the 31 participants 20 were female and 11 were male. Nearly 20% of participants were chlamydia positive and 29% of participants attended STD education. In comparison to the fall academic year where chlamydia positivity rates were 32%. **Conclusion:** A reduction in chlamydia cases was observed after introduction of an STD educational program and fast track STD screening, however more research and efforts are needed to decrease chlamydia cases on campus overall and increase the use of mental health services.

Keywords: Chlamydia, College Students, STD, Sexual Health

Developing and Implementing a Sexually Transmitted Disease Screening Program to Decrease Chlamydia Cases on a College Campus

Chlamydia is one of the most common sexually transmitted diseases (STDs) among young adults in the United States of America (USA). According to the Centers for Disease Control and Prevention, (CDC), Chlamydia is also one of the most frequently reported STDs, (CDC, 2022). In the year 2020, there was an estimated 1.6 million cases of chlamydia reported to the CDC. Rates of cases are highest among adolescents and young adults between the ages of 15 and 24 (CDC, 2020). Chlamydia has also caused a financial burden in the United States. It costs the healthcare system about 690 million dollars annually (CDC, 2021). Given these high statistical cases and cost to the healthcare system, it is imperative to implement strategies to reduce the number of chlamydia cases.

Chlamydia has also disproportionately impacted minority groups. It is most prevalent in lower socioeconomic groups, especially in minorities, both young males and females (Kelly & Stephens, 2004). Given this statistical data, more interventions should be tailored to meet the needs of this population. It is also important to recognize that chlamydia can have a huge impact on overall health especially in women. Chlamydia can lead to poor health outcomes because it often goes unnoticed. It can lead to infertility, pre-term births, and an increased risk of contracting or spreading Human Immunodeficiency Virus (HIV) (CDC, 2022). Research has not identified the driving factors of why Chlamydia cases continue to rise among young adults. Evidence suggests that the decreased use of condoms, the decline in state and federal STD programs, the rise in antibiotic resistance, the increase in sexual dating applications, and decreased screening are major factors that contribute to the increase in cases (Keller, 2020). Given this current research, and the negative impact chlamydia has on young adults, more efforts should be directed towards STD prevention strategies to improve patient outcomes.

Prevention strategies are needed to decrease the incidence of chlamydia in this vulnerable age group. Current screening guidelines set forth by the United States Preventive Services Task Force (USPSTF) make suggestions for practice by grades of recommendation. A grade A recommendation according to the USPSTF, strongly recommends that clinicians provide service to patients. Grade B is a recommended service with high certainty that the benefits and the service should be provided (USPSTF, 2020). A grade C makes no recommendation for or against service. Grade D recommends against routinely providing service to asymptomatic patients. Lastly, a Grade I recommendation concludes that there is insufficient evidence to recommend for or against routinely providing service (USPSTF, 2020).

Current guidelines set forth by the USPSTF state that all sexually active women ages 24 years or younger should be screened for STDs which is a grade B recommendation. It also recommends that all sexually active adolescents and young adults who are at increased risk of STDs have behavioral health counseling, which is also a grade B recommendation (USPSTF, 2020). Even with the current recommendations, research has shown that chlamydia cases continue to rise among young adults, and disparities persist among ethnic minority groups (CDC, 2022). Limited evidence to examine this issue also contributes to the negative impact of chlamydia amongst this age group. This research evidence is important to this project as there has been an increased number of chlamydia cases on a college campus among young adults. Developing and implementing interventions to help combat this clinical issue was important to help decrease chlamydia cases on campus overall.

Problem Statement

Over the past academic school year, there has been a significant rise in the number of chlamydia cases on a college campus, particularly among young adults aged 18-25. This increase poses a major public health concern, as chlamydia often remains asymptomatic, leading to undiagnosed and untreated cases that can result in serious health complications. The spike in infections suggests a need to address potential gaps in sexual education, access to preventative measures, and regular screening services for students. Additionally, the increase may be related to behavioral health factors, such as lack of condom use, and lack of awareness about asymptomatic transmission, which contributes to the spread of infection within the college community. Addressing this clinical issue requires a comprehensive approach, which includes increasing the knowledge of chlamydia on campus, improving access to screening and treatment services, and outreach to encourage healthy sexual behaviors. Without intervention, the increase in the incidence of chlamydia can lead to chlamydia spreading on campus and into the surrounding community. To examine further the need to address this clinical issue it is important to discuss the background of the problem.

Background of Problem

There were several problems identified through STD screening at a college campus wellness center at a predominantly black college (PBC) in Chicago, Illinois. The problems identified were a large increase in the number of positive chlamydia cases on campus, an increased spread of chlamydia, and an increased amount of re-infection after successful treatment of positive chlamydia cases. Documentation of this problem was examined by performing a preliminary analysis. This was reviewed by conducting an initial chart review,

which revealed a high number of positive chlamydia cases among young adults ages 18-25. Interviews with patients who have positive partner-confirmed cases were also a source. Through interviews, promiscuous sexual behaviors were identified.

A preliminary review of data was collected from September 2021 to May 2022 to support findings. Chart reviews revealed that there was a total of 87 (58%) positive cases of chlamydia on campus. Most of the confirmed cases came from students residing in the on-campus residence halls. Of the 87 cases males accounted for 48 (55%) of the cases and females accounted for 39 (45%), Of the 87 cases, 11(13%) cases were discovered during routine screening, 8 (9%) displayed symptoms, and 68 (78%) were encouraged to get testing based on reported high-risk sexual behaviors. The ethnicities screened were Hispanics, Caucasians, Asians, and African Americans. Of the 150 patients screened, African American students accounted for all 87 cases. This continues to support that minority groups are impacted at higher rates and can evolve into a public health problem.

Chlamydia is a significant public health problem that involves multiple providers. This disease may have a major impact on the community and potentially proliferate beyond the college campus. According to the Illinois Department of Public Health (IDPH), in 2021, approximately 16,000 chlamydia cases were reported in Cook County (IDPH, 2021). Proliferation into the community will increase cases and may become a public health issue. This issue will increase clinic visits for Advanced Practice Nurses (APNs), as well as the use of resources. It will also require close follow-up for the test of cure, which according to the CDC is repeat testing 4 weeks after completing therapy to help detect if the treatment for the initial infection failed (CDC, 2021). This can lead to increased healthcare costs. Untreated chlamydia can lead to poor patient health outcomes. In women, untreated chlamydia can lead to pelvic

inflammatory disease (PID). This can lead to infertility, chronic pelvic pain, and continued spread of the disease (CDC, 2021). Chlamydia can also be a gateway to other serious diseases such as Human Immunodeficiency Virus (HIV). This has a serious impact on healthcare utilization and the demand for APNs' management of chlamydia. Advanced Practice Nurses play a pivotal role in healthcare. They are responsible for the treatment, diagnosis, prevention, and management of diseases such as chlamydia (American Nurses Association, 2024). The demand for APNs will increase if the cases increase.

PICO

A clinical question was developed utilizing the Population, Intervention, Comparison, and Outcomes approach (PICO). The purpose of the PICO question is to strategically develop and frame a research question (Aslam, 2010). This research question is the initial step to conducting valid and valuable research. The specialized framework uses patient/ population, intervention, comparison and outcome (Eldawlatly et al, 2018). **Population:** College students ages 18-25, including all genders. **Intervention:** Introduction of a new STD screening program that implements a fast-track STD screening, to increase the uptake after testing, in combination with increasing awareness on campus through educational outreach, distribution of resources, and referral to behavioral health. **Comparison:** Compared to current recommendations and evidence-based guidelines by the United States Preventive Services Task Force to regularly screen sexually active women 24 years of age or younger. The USPSTF also recommends screening women who are 25 years or older at increased risk for infection screening. **Outcomes:** The goal was to implement a new screening program to detect chlamydia cases on a college campus, increase knowledge of the disease, and increase access to mental health services to decrease the spread and incidence of chlamydia on a university campus.

Needs Assessment

Conducting a needs assessment was important to this project in identifying any current gaps in practice. A needs assessment is a systematic process used to determine if there are needs or gaps between current and desired outcomes (Asana, 2024). A needs assessment involves three phases: pre-assessment, assessment, and post-assessment. A preliminary needs assessment took place at a wellness center on a university campus in Chicago, Illinois, where students and staff members are serviced at the university. At the wellness center, students are provided medical services which include but are not limited to preventative testing, health screenings, and health education. Most of the population that seeks care at this facility are young adults. The services offered are free to students if they attend the university. This needs assessment focused on the needs of the college students particularly those who were mostly affected, and those were young adults aged 18-25. The goal was to assess and incorporate evidence-based strategies into practice to decrease the prevalence of chlamydia cases affecting these individuals. Strategic planning efforts involved a combination of EBP strategies along with a proposed action plan for potential chlamydia outbreaks on campus in the future.

Pre-Assessment: A meeting was conducted with the key stakeholders of this project to identify and discuss current gaps in practice. The scope of this phase was strategic. It involved establishing goals and objectives for the university and the community that it served. Utilizing different tools and techniques to implement a change had a positive outcome for these patients. During the preliminary assessment, data collection was extracted from 150 direct patient charts through an Electronic Medical Record (EMR) that tested positive for chlamydia from September 2021 until May 2022. Preliminary data on the number of chlamydia cases were from totaled 87 positive cases.

Assessment: The purpose of the assessment phase is to implement the evaluation in a manner that produces justifiable information to make informed decisions (Watkins et al, 2011). During the 2021 academic school year, a review of current clinical guidelines to address screening for STDs amongst the youth was considered. Screening recommendations set forth by the CDC are to screen for chlamydia in women who are sexually active and under 25 years of age, screen sexually active women 25 years and older at increased risk and retest approximately three months after treatment (CDC, 2024). A questionnaire was discussed in the first team meeting to potentially gain a better insight into student perspectives of sexually transmitted diseases. The purpose of a questionnaire in research is to get the most relevant information in a reliable manner (Taherdoost, 2018). A tool was adapted from the CDC called A Guide to Taking a Sexual History. This tool offered a guide for discussing sexual health with patients. It involved a series of questions that were guided by the Five "Ps", which was adapted into a questionnaire form to assess the current and past sexual history of young adults (see Appendix C). The goal of the 5Ps was to improve the patient's health (CDC, 2021). The 5Ps stand for partners, practices, protection from STDs, history of STDs, and pregnancy intention (CDC, 2021). The tool was intended to assess the sexual health history to help identify the services that the patient needs and if they qualified for the fast-track testing service. This type of risk assessment tool was beneficial for the students at the university and a quick and easy tool to help assess the need for STD screening and the use of fast-track STD screening. It identifies risky sexual behaviors and signals providers to screen for STDs.

The preliminary data on the number of chlamydia cases were reviewed with the team in June of 2022 totaling 87 positive cases. A Strength, Weakness, Opportunities, and Threats (SWOT) analysis was also performed. The purpose of the SWOT analysis is to identify,

prioritize, and organize the strengths, weaknesses, opportunities, and threats that influence project planning, development, implantation, and evaluation (Watkins et al, 2011).

Table 1. SWOT Analysis for Needs Assessment

Strengths	**Weaknesses**
• Full support of leadership at the university • Affiliation with the university counseling department	• Lack of education presented in freshmen orientation. • Lack of resources in college dormitory outside of wellness center hours
Opportunities	**Threats**
• Support from CDPH, supplies condoms, educational material, monthly rapid HIV screening free of charge, Free STD screening inclusive of chlamydia	• Chicago Department of Public Health integrating new online reporting. Paper reports are still accepted. • Delayed results and contact tracing. • Stigma related to STIs. • Poor follow-up with test of cure

Note. An analysis describing organizational needs, strengths, weaknesses, opportunities, and threats.

The results of the findings were discussed, and a proposal for integrating sexual education and awareness services was recommended on campus. It Included screening for and testing for Chlamydia. Recommendations were that this integration be incorporated into the mandatory orientation before classes start. If resources are the issue, condoms, and educational materials will be distributed to students free of cost.

Post Assessment: After the preliminary needs assessment for this age group, recommendations are to assess the current knowledge base about chlamydia. The current

recommendation according to the CDC, is screening all sexually active women under 25 years of age and annually for all sexually active young adults (CDC, 2022). An action plan was discussed based on these recommendations. To support this clinical issue a follow-up chart review was performed to assess the number of chlamydia cases on campus. From the fall academic school year, September 2023-November 2023. In total, 50 charts were reviewed. The positive chlamydia cases totaled 16 positive cases over a two-month period. The chlamydia positivity rate from that sample for this period was 32%. Evidence revealed that the problem stemmed from both males and females, students who had multiple sexual partners, and students who refused to use condoms. From a provider standpoint, this issue raised many concerns for the safety and well-being of this population. There was a need for change and the implementation of evidenced-based strategies to reduce the incidence of chlamydia amongst this population of students. The findings were discussed with stakeholders. Key stakeholders in the needs assessment are outlined. (Appendix B).

Based on the data collected and the projected positivity rate, an action plan strategy to reduce and prevent chlamydia on the university campus was discussed. The goal was to provide resources so students could be more knowledgeable about the disease and ways to prevent it. This was communicated by raising awareness on campus, which was achieved by hosting an educational outreach event about chlamydia and other STDs. One of the major issues identified was the lack of education on chlamydia. Many students lacked basic knowledge about chlamydia, such as signs and symptoms and curability of chlamydia. Incorporating education on chlamydia and other STDs helped students identify knowledge gaps and gave them the tools needed to help prevent chlamydia using contraception and early detection. To help identify the

latest evidence-based research to develop interventions for this project, a review of literature was conducted.

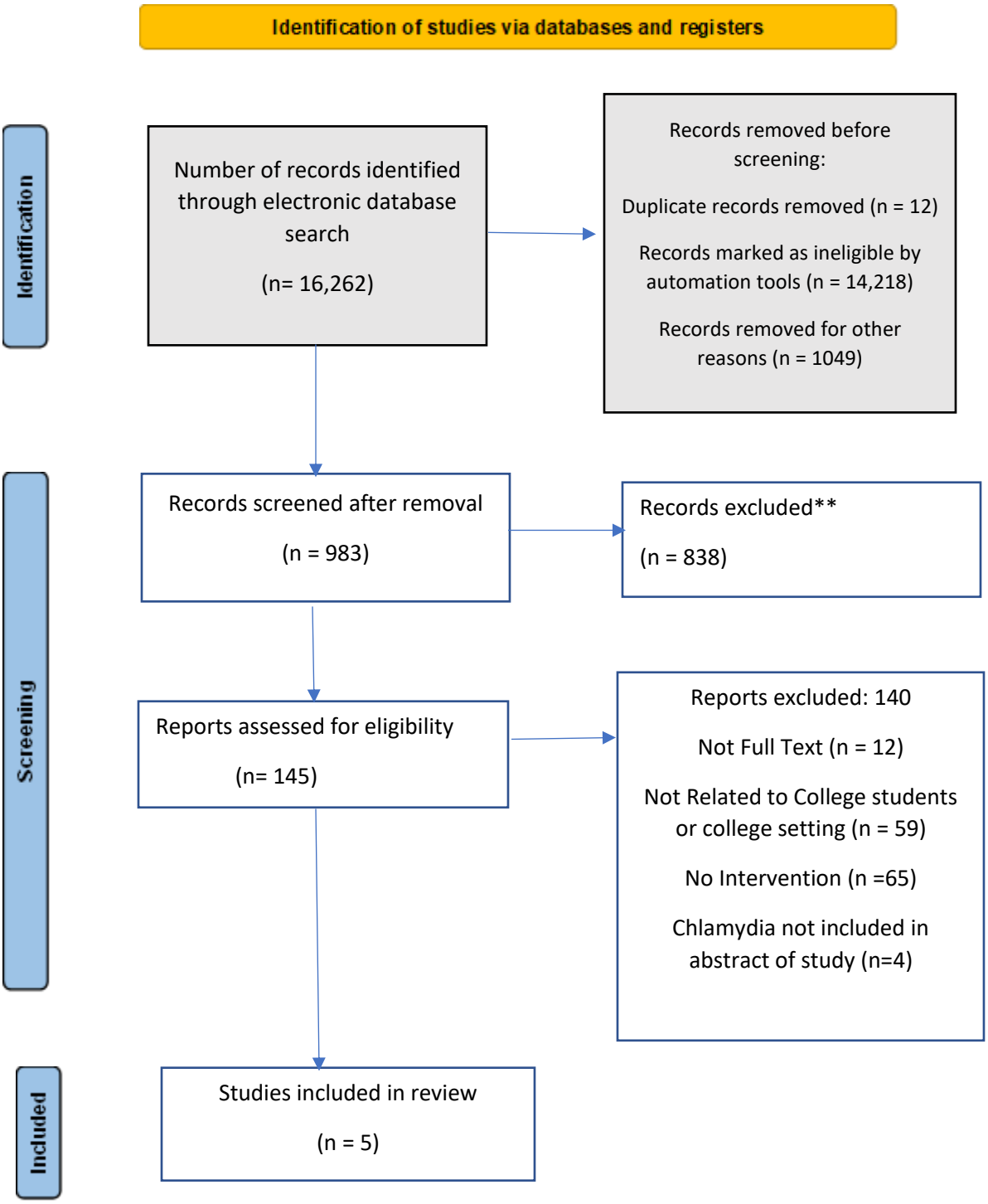
Integrative Review of Literature

The purpose of the integrative review of the literature was to synthesize current evidence-based interventions to support project decision-making and examine prevention strategies to decrease chlamydia among college students. Utilizing experimental and non-experimental research and reviewing current and past theories helps identify gaps in literature and defines and analyzes methodological issues (University of Houston, 2023). Quantitative and qualitative data were considered.

Search Strategy

Search databases included PubMed, Science Direct, and Cumulative Index to Nursing and Allied Health Literature (CINAHL). Electronic databases searched can be found in (Appendix A). Search words and terms utilized were “Chlamydia and College Students”, “Chlamydia Prevention in College Students”, “Chlamydia in Universities”, and “Chlamydia and College.” Abstracts were reviewed rigorously to establish whether research articles were relevant. Full-text articles were only considered for review of relevancy and to support review. Reference lists were also reviewed in the Integrative review. The search results identified 16, 262 articles. After records were removed due to inclusion, exclusion, duplicates, and ineligibility by automation tools, 983 articles were considered. A flow diagram to support article selection is displayed in Table 2.

Table 2.



Note. A Flow diagram with identification of studies to support article selection.

Criteria for considering studies

The articles sought were aimed at chlamydia prevention and or reduction on college campuses. This review primarily evaluated studies that focused on chlamydia cases at a campus university or a prevention program to reduce the incidence of chlamydia. For the best evidence, randomized control trials (RCTs) were sought; however, due to limited research and evidence qualitative studies were also considered. For the study selection articles, were independently reviewed written in the English language and contained full text for inclusion criteria. The population being investigated were young college adults ages 18-25 years old. Given limited data on young adults 18-25, articles of other ages discussed if the range was included. Explanation of inclusion and exclusion can be found in table 3.

Table 3.

Inclusion Criteria	Exclusion Criteria
Young adults ages 18-25 must be included in age range	Ages 18–25-year-old not included in the study, adolescents, middle-aged adults, elderly
Chlamydia must be included in any study regarding STDs	Chlamydia not included in article discussion, unrelated
Must be on a college or university campus	Any other setting besides a college university or campus
Male, female, or transgender	Not in the English language
Only English language studies	Not full-text articles
Intervention based studies	Has no intervention or prevention suggestions
Full-text articles	Payment needed to retrieve the full text
Articles published from 1960- 2024	Articles published before the 1960s and after 2024

Note. This table displays the inclusion and exclusion criteria to identify articles eligible for review.

Results

Of the 983 articles reviewed, 145 were assessed for eligibility. Excluded were 140 of the articles because they did not meet the inclusion criteria, Table 3. A lack of evidence existed

regarding chlamydia prevention on college campuses. No clear interventions have been researched and evidence-based practices were not clearly identified. Of the five articles identified that looked at intervention-based strategies to prevent and decrease chlamydia on college campuses, all stated that more research is needed. One study involved a theory-based sexually transmitted disease prevention program for female college students. The post-intervention assessment demonstrated increased STD knowledge and reductions in the number of sexual partners. This showed a positive correlation between increased knowledge and changed behavior (Jaworski & Carey, 2001). Another RCT focused on screening as a preventative method. Incorporating regular chlamydial screening in college-aged adults proved to be effective. Testing identified positive cases and stopped the spread of chlamydia (Hennrikus et al, 2009).

Another intervention discovered in the article reviewed was a campaign geared towards college students to raise awareness about STDs and testing. This campaign raised awareness and encouraged STD testing on a college campus. This technological approach, as well as increased awareness, had a positive response on campus. This intervention increased the number of student's knowledge regarding sexually transmitted diseases and decreased the number of cases on campus (Habel et al, 2015). Lastly, another assessed the knowledge and attitudes about STDs. The study showed that college-age students lacked the necessary knowledge needed to avoid a sexually transmitted disease (Folasayo et al. 2017). Overall, the only study that utilized evidence-based strategies was the one that highlighted screening as an intervention.

Currently, the United States Preventative Services Task Force (USPSTF) recommends chlamydia and gonorrhea screening for all sexually active women, including pregnant individuals. Providers should screen for chlamydia in women 24 years or younger and in women

25 years or older at risk for infection (USPSTF, 2021). The CDC has similar screening guidelines that recommend screening sexually active women routinely under the age of 25 years old. It also recommends screening sexually active women 25 years and older at high risk (CDC, 2024).

An updated search from 2023 yielded similar results. After reviewing the current literature, a new article investigated the impact of stigma and accessibility of STD testing amongst college students. An integrative review was performed on the potential use of STD/HIV self-test kits for college students. This was relevant to the goal of decreasing chlamydia on campus because self-testing kits may help identify chlamydia early and promote screening. The research questions developed for this study were do self-test kits increase testing? How do stigma and lack of access impact STD screening amongst college students? The results of this study again expressed limited research and potential barriers such as lack of insurance coverage, embarrassment and shame, poor testing habits, and lack of transportation as roadblocks for college students (Reeves et al, 2023). Given this new information, when research becomes more available, self-test kits for chlamydia and other STDs may be of benefit for this population. After an exhausting search of evidence, it is apparent that more research is warranted in this area and evidence-based strategies need to be implemented. Latest evidence-based guidelines to support this project was incorporated due to limited findings and can be found in Appendix A.

Quality and Assessment of Grading

The articles that were selected for the review were assessed utilizing the New Joanna Briggs Institute (JBI) levels of evidence as a guide for evidence effectiveness and

meaningfulness. The JBI grades of recommendation were also utilized. The quality and assessment grading are displayed in Table 4.

Table 4. The Joanna Briggs Institute Levels of Evidence

Levels of Evidence
Evidence for Effectiveness **Level 1 – Experimental Designs** Level 1.a- Systematic review of Randomized Controlled Trials (RCTs) Level 1.b Systematic review of RCTs and other study designs Level 1.c- RCT Level 1.d- Pseudo-RCTs **Level 2- Quasi-Experimental Designs** Level 2.a-Systematic review of quasi-experimental studies Level 2.b- Systematic review of quasi experimental and lower study designs Level 2.c- Quasi-Experimental prospectively controlled study Level 2.d- Pre-test-posttest or historic/retrospective control group study **Level 3- Observational- Analytic Designs** Level 3.a- Systematic review of comparable cohort studies Level 3.b- Systematic review of comparable cohort and other lower study designs Level 3.c- Cohort study with control group Level 3.d- Case-controlled study Level 3.e-Observational study without a control group **Level 4- Observational Designs- Descriptive Studies** Level 4.a- Systematic review of descriptive studies Level 4.b- Cross-sectional study Level 4.c- Case series **Level 5- Expert Opinion** Level 5.a- Systematic review of expert opinion Level 5.b- Expert consensus Level 5.c- Bench research/single expert opinion
Levels of Meaningfulness **Level 1-** Qualitative or mixed-methods systematic review **Level 2-** Qualitative or mixed-methods synthesis **Level 3-** Single qualitative study **Level 4-** Systematic review of expert opinion **Level 5-** Expert opinion

Note. Supporting Document for the Joanna Briggs Institute Levels of Evidence and Grades of Recommendation, by The Joanna Briggs Institute (2014), The University of Adelaide. Adapted with permission.

The JBI Grades of Recommendations was utilized to identify the best evidence-based strategies and interventions for decreasing chlamydia amongst the college student population. The JBI has two grades of recommendations identified in Table 5.

Table 5. The Joanna Briggs Institute Grades of Recommendation

JBI Grades of Recommendation
Grade A- "strong" recommendation for a certain health strategy. (1) The desirable effect outweighs the undesirable effect of strategy. (2) there is evidence of adequate quality supporting use. (3) there is a benefit or no impact on resource use. (4) values, preferences, and patient experience considered
Grade B- A "weak" recommendation for a certain health strategy (1) desirable effects appear to outweigh undesirable effects of strategy. (2) Where there is evidence supporting use, although it may not be high quality. (3) there is no benefit or impact or minimal impact on resources used. (4) values, preferences, and the patient experience may or may not have been considered.

Note. Supporting Document for the Joanna Briggs Institute Levels of Evidence and Grades of Recommendation, by The Joanna Briggs Institute (2014), The University of Adelaide. Adapted with permission.

Literature Synthesis

Existing evidence shows the lack of data to support evidence-based strategies and interventions to reduce chlamydia on college campuses. Key issues that were identified within this review were lack of knowledge on sexually transmitted diseases from students, lack of resources, and lack of screening amongst this population. A comprehensive prevention program is needed, as well as an action plan campus-wide to decrease the rate of chlamydia cases on a university campus. According to an article reviewed, college campuses must develop campus-wide policies that address STI prevention. This article suggests that there should be referral

systems to connect students to care in the community, such as local health departments and Federally Qualified Health Centers (Habel et al, 2016). Giving these students the knowledge and resources to help with this disparity is crucial. This can be done by increasing awareness through awareness programs.

Based on the current screening recommendations from the CDC (2022) to screen all sexually active women under 25 years of age and annually for all sexually active young adults recommendation, given the age group, a quality improvement plan was implemented. This program allowed for fast-track STI screening for asymptomatic patients. Individuals bypassed the appointment process, left a specimen and were contacted if further follow-up is needed. If an individual is symptomatic, they could access same-day services which included screening and treatment if necessary. A pre-existing guide adapted from the CDC on taking sexual history was adapted and given to patients before STD fast-track appointments. This serves as a guide to assess overall sexual health and can be utilized as a quick reference resource when reviewing a patient's sexual history. This questionnaire can also be used to assess high-risk sexual behaviors. This screening tool was used as a guide to recommend further testing or counseling on sexual behaviors. Increased screening can be implemented by screening according to current guidelines and new screening methods. Development or adaptation of a pre-existing screening tool to assess high-risk sexual behavior should also be deployed to signal the provider to test given a high score. Many factors should be considered given this population, and attitudes and beliefs about sexually transmitted diseases should not be ignored. The effects of a brief theory-based intervention seemed to be the most effective. Implementing this strategy into incoming freshmen orientation may be of benefit by increasing knowledge of sexually transmitted diseases through

educational videos. This may lead to a reduction in high-risk sexual behaviors and a decrease in the incidence of STDs, including chlamydia.

Discussion

This integrative review was conducted to identify current evidence-based strategies and interventions to help decrease and prevent chlamydia among college students. The goal was to gather enough evidence to strategize and implement a sexually transmitted disease prevention program for college students. The evidence gathered shows a lack of research and resources available to help implement strategies to decrease the incidence of chlamydia among these young adults. The evidence continues to push for more screening and increasing knowledge. However, there is a lack of interventional programs for at-risk youth. While current guidelines point to screening and behavioral health counseling as an intervention, more research-based interventions should be implemented as numbers continue to rise. The answer to this PICO question remains at large as more data is needed to support the implementation of a successful prevention program for college students.

Many gaps existed while conducting this review as data is not current and more evidence-based interventions have not been implemented. Due to time constraints and limitations on the review of articles independent, information extracted from the literature was limited. The studies derived for this review included randomized control trials, Quasi-experimental designs, observational-descriptive studies, and quantitative and qualitative studies. Intervention recommendations included increasing the knowledge of chlamydia among college students, routine screening during visits, campaigns geared towards prevention and awareness, and behavioral health interventions. While collectively, these may be of benefit in reducing the

incidence of chlamydia, more studies should be performed to assess the likelihood that these interventions will decrease chlamydia amongst college students. As chlamydia continues to cost the healthcare system millions of dollars, financial support should be put towards resources for this population to decrease the incidence as well. To effectively guide this capstone project and deepen the understanding of behavior, it is essential to establish a clear conceptual framework.

Conceptual Framework

The framework that guided this project was the Health Belief Model (HBM). This model was developed in the 1950s by social psychologists Rosenstock, Hochbaum, and Kegeles. Originally, this model was based on four constructs. The four constructs were perceived susceptibility, perceived seriousness, perceived benefits of taking action, and the barriers to taking action and cues to action (Rosenstock, 1974). Later, this model would evolve to six constructs to address public health concerns. This psychological model attempts to explain and make predictions on health behaviors based on the attitudes and beliefs of the individual related to health (LaMorte, 2019). It is based on the individual's beliefs in the personal threat or illness. The belief in the effectiveness of recommended health behavior action will predict the likelihood a person will adopt the behavior (LaMorte, 2019).

This model is also based on the understanding that a person will take a health-related action if the person; feels a negative health condition such as chlamydia can be avoided. The model explains that if an individual has a positive outcome from the recommended action, they will avoid the negative condition. For example, as it relates to this project using condoms or abstaining from sexual intercourse to avoid STDs. Lastly, this model explains that if an

individual believes that he can achieve the health behavior change, then they will have better outcomes.

For this project, the six constructs were adapted, which included the original four constructs and additional constructs that were later added. The six constructs are perceived susceptibility, perceived severity, perceived benefits, perceived barriers, cue to action, and self-efficacy. To help get a better understanding of why the occurrence of chlamydia was increasing it was important to understand the person's perception of illness and disease. Integrating this model into this project helped to gain a better understanding of how patients perceived contracting chlamydia and what interventions could be implemented to decrease the number of cases on campus. The first element is perceived susceptibility, the person's perception of disease or illness (LaMorte, 2019). During the educational phase of this project, the perception of students was that they did not use condoms because they felt their sexual partner was clean. The participants also did not perceive chlamydia as a threat because they never had an STD in the past and had trust in their partners. Some participants partook in high-risk sexual behaviors because they were not knowledgeable about the different STDs they could contract. The perception of chlamydia not being a threat led to high-risk sexual behaviors.

The second element is perceived severity, meaning feelings on the seriousness of having a sexually transmitted disease and the dangers if left untreated (LaMorte, 2019). The young adults who participated in the fast-track STD screening and/or educational intervention were unclear of the implications surrounding chlamydia (PID, infertility). During the STD educational event, some participants saw chlamydia as less severe than other STDs because it was curable. The third element is perceived benefits and attitude, which relate to the effectiveness of action (LaMorte, 2019). Some individuals had peace of mind knowing there was a cure for chlamydia.

With this knowledge, they continued to partake in risky sexual behaviors. This was observed during the discussion after the educational intervention. It was also evident after observing some of the responses to the sexual health questionnaire, which included questions related to past STDs as well as condom use.

The fourth element is perceived barriers which relate to the obstacles to performing health action (LaMorte, 2019). Barriers identified during fast-track screening and STD educational outreach were that students faced fear of testing, fear of results, lack of access to care, and stigma related to having an STD. Because of these barriers, students endorsed that they did not seek testing which can lead to negative outcomes. The fifth element is cues to action. This is related to the stimulus needed to trigger decision-making (LaMorte, 2019). Individuals who sought testing in the past were those that were exposed to STDs, or experienced symptoms related to STDs. Regarding this project, a cue to action was the education stimulus that informed students of chlamydia, modes of transmission, and symptoms. Students viewed this information as beneficial and decided to seek testing to know their current health status as it related to their sexual behaviors. Lastly, self-efficacy relates to the level of confidence in an individual's ability to perform the behavior (Lamorte, 2019). Barriers to this may be related to young adults not knowing how to find a testing site, they lack the necessary education about STDs, or they do not know how to start the conversation with primary care providers due to the stigma related to having an STD.

There were limitations to this theory. Students who did not attend education and only participated in the fast-track STD screening lacked knowledge about chlamydia and other STDs. This can have a negative impact on the individual affected and the community. Lack of knowledge leads young adults to poor decision making, such as partaking in high-risk sexual

behaviors not knowing the consequences of such behavior. This theory also did not account for participants' attitudes or beliefs that guided their sexual behaviors. Some young adults who participated in the educational intervention did not see chlamydia as serious because there is a cure. Some attitudes towards STDs and condom use did not change pre- and post-educational intervention. These views and attitudes can contribute to the negative impact of this disease on the campus community.

Lastly, this model does not consider habitual behaviors (Lamorte, 2019). Some of the students just did not believe in condom use. If they had an STD in the past, they admitted that they sought treatment and continued high-risk sexual behaviors afterward. Although there are limitations to this theory, this theory can positively guide practice. It promotes prevention and helps patients take accountability for their health. If students can change their perception of STDs, they may have the ability to avoid high-risk sexual behaviors that lead to them. This theory can help healthcare providers guide care by increasing the awareness needed to enable patients to make better decisions regarding their sexual health. This theory helps build upon the knowledge of the Advanced Practice Nurse (APN), and the patient by providing education about chlamydia, prevention, diagnosis, and treatment.

Purpose of Project

The purpose of this project was to determine if implementing a quality improvement program on a university campus utilizing a holistic approach will help decrease chlamydia cases. The program focused on fast-track STD screening for young adults between the ages of 18-25. The procedures involved bypassing the need to schedule an appointment, as well as seeing a provider, and the ability to walk-in and receive fast-track STD screening services. It also

involved increasing knowledge of STDs through educational interventions and addressing mental health issues using counseling services. This project integrated the latest evidence-based guidelines into current practice to decrease the number of chlamydia cases on campus. These interventions were important to this specific population because the number of chlamydia cases was on the rise and negatively impacted young adults.

Implementing this project program was the first step to creating an interventional-based method to decrease the spread, incidence, and number of cases on a university campus. The aim was to introduce a free fast-track STD screening program on a college campus to increase testing uptake. Increasing access to care would promote the identification and treatment of chlamydia on campus. This project also aimed to increase the knowledge of STDs and decrease the stigma related to STDs. Many participants lacked basic knowledge about STDs and healthy sexual behaviors. They also were not equipped with the necessary resources for prevention.

Measurable Goals and Objectives

The overall goal of this capstone project was to decrease the number of chlamydia cases on a university campus. Projected short-term goals included reducing the incidence and spread of chlamydia by 50%. Long-term goals were to see a decrease in chlamydia cases on campus to less than 25% per academic school year. Three main goals guided this project. The goals were to increase knowledge of chlamydia on campus through an STD educational intervention, increase access to care through a new STD screening program, and increase access to mental health services for behavioral health counseling for individuals who were identified as high risk for chlamydia and other STDs.

The first goal was to increase the knowledge of chlamydia and other STDs on campus through an STD educational awareness event. The objective was to increase knowledge about chlamydia so students could make more conscious choices to prevent contracting the disease. To help meet this objective, an STD educational event took place. This event took place on January 23, 2024. This educational intervention included an information session as well as a pre-and post- STD educational survey. These surveys assessed the attitudes of students towards STDs pre- and post-educational intervention.

The educational intervention consisted of two 30-minute informational sessions which included a 15-minute STD educational awareness PowerPoint presentation. The PowerPoint presentation included information on chlamydia and other STDs. This included statistical data such as prevalence and incidence in the United States, retrieved from the CDC. It also included information on the signs and symptoms of STDs, treatment, prevention, and curability. This event was open to the campus community. However, the students invited to participate in the survey met the inclusion criteria. The inclusion criteria were that students had to be 18-25 years old and inclusive of all genders. Students who opted-in to participate in the educational intervention signed a consent form and participated in a pre-and post-STD educational survey. Students were able to ask questions regarding healthy sexual behaviors and gain knowledge of all STDs.

The next goal was to increase access to healthcare. This goal was achieved by implementing the fast-track STD screening program on campus so that students could walk in and access to services on demand. The objective was to expand healthcare services and bridge barriers students may face when accessing care. These barriers included the inability to get an appointment with their primary care provider within a prompt timeframe, no insurance to cover

the cost of the visit and convenience of location. With the fast-track program, students could access the services needed most without the hassle of insurance and additional costs. Improving access to screening helped to increase the uptake of testing on campus. Compared to the fall semester of 2023 and the spring semester of 2024, there was a slight decline in chlamydia cases on campus. For the fall academic school year, chlamydia cases on campus were at 32%, while the spring chlamydia case rate was 20%. This shows a 12% reduction rate in chlamydia cases. One can draw inferences that increasing access to care and increasing awareness impacted the reduction of chlamydia cases on campus.

Increasing access to mental health services was another important goal for this project. The objective was to refer high-risk sexually active youth to the counseling department for behavioral health services. This service was also free of charge. Following the USPSTF guidelines, referring patients with high-risk sexual behaviors is a grade B recommendation. As previously stated, a Grade B is a recommended service with high certainty that the benefits and the service should be provided (USPSTF, 2020). Following these recommendations, students considered high-risk were referred to the counseling department.

Methodology

Project Design

This project design included both descriptive/ and quasi-experimental, pre-test and post-test. The quasi-experimental design will be used to evaluate the interventions of both the STD educational intervention as well as the fast-track STD screening. This will allow for an estimation of the impact of the intervention and a comparison of outcomes between participants. A Pre-test and post-test STD educational survey will be utilized to measure the outcomes/

attitudes of participants before and after STD educational intervention. Descriptive data will be used to analyze the characteristics of the population. This data will include demographical data such age, race, Hispanic or Latino Origin, and year in college. Quantitative and qualitative data will be used to document trends and patterns to investigate the impact of educational and screening interventions on decreasing the number of chlamydia cases on a college campus.

Sample

The target population for this project was young adults between the ages of 18-25, inclusive of all genders and races. The sample size was from three groups. The sample size for the fast-track screening program was 31. The sample size for the participants who participated in the STD educational program was 31. The sample size for those who participated in both programs was nine.

Setting

The setting for this project is the wellness center on a university campus. The wellness center is a place where students can seek medical care while they are on campus. Services provided to the students include but are not limited to STD screening, immunization compliance, sick visits, and school physicals. Students can access these services free of charge. The university where this wellness center is located is in an urban setting. The student population primarily serviced at the wellness center are young adults who are African American. These students are from the surrounding areas or reside in the residence halls on campus. The surrounding community is also urban. The decision to choose this university was based on the increased number of chlamydia cases occurring on campus.

Statistical Methods

A descriptive and quasi-experimental, pretest and posttest group design was utilized for this quality improvement project. The statistical tests and procedures included Excel spreadsheets and IBM statistical package for the social science (SPSS) version 23. The Shapiro-Wilk test was used to evaluate the normality of the distribution. The Wilcoxon Signed Rank Test was used to examine the Pre-and Post STD survey, fast-track STD screening, and Sexual Health Questionnaire. The Fisher's Exact test was used to determine statistical significance.

Protection of Human Rights

Ethical considerations were considered for the duration of this project. The procedures conducted strictly adhered to the ethical principles outlined in the Belmont Report (U.S. Department of Health, Education, and Welfare, 1979). All participants were treated with dignity and respect, and their autonomy was protected throughout this project. Prior to participation in any surveys or screenings, participants were provided a clear and comprehensive explanation of the project's purpose, procedures, and benefits. Participants were required to give informed consent in writing, indicating their voluntary agreement to participate. They were informed that they could withdraw from the program without penalty.

The participant's privacy was protected, and all data was kept confidential. Personal identifiers were removed from the data, and the published results did not include any information that could lead to the identification of any of the participants. Data was secured and stored only by authorized personnel. Every effort was made to minimize any potential risks to participants. This included psychological, emotional and physical risks. Participation in the activities within this project was strictly voluntary and participants were informed of their right to withdraw at any point without any negative consequences. This study was reviewed and approved by both

University Institutional Review Boards (IRBs) to ensure it meets ethical standards and that the rights and welfare of the participants are protected.

Data Collection Tools

There were multiple tools utilized to support data collection for this project. For the STD educational event a pre-and post STD educational survey was utilized (Appendix D). This survey was comprised of a series of questions that assessed the attitudes of participants toward STDs before and after the STD educational intervention. The project leader created this survey and assessed it for validity and reliability. The tool was reviewed by two Family Nurse Practitioners, with a doctorate degree in nursing and specialize in preventative care. Both commented that the survey contained appropriate statements that related to the clinical issue, and the content made sense as it relates to the project. It aligned closely with theory-based interventions such as the health belief model. Utilizing the behavioral change theories helped support and develop interventions to help understand behaviors (Heath et al., 2015). The pre-and post STD survey contained questions to help support that introducing an educational stimulus about chlamydia and other STDs could shift the attitudes of participants towards STDs. This survey also aligned with the recommendations from the CDC to illicit questions regarding sexual health to effectively assess an individual's sexual history (CDC, 2021).

For reliability, the test-retest method was utilized. Three randomly selected students were asked to participate and return for the posttest. Results showed that when students took the survey the first time, the responses did not change for the post survey. This shows the reliability in this tool. Students were asked to answer questions on a Likert scale, which included Strongly Agree, Agree, Strongly Disagree, Disagree, Neither Disagree nor Agree. This survey helped to

support the project by assessing the attitudes of young adults towards STDs before and after STD educational intervention.

The next tool utilized was adapted from the Centers for Disease Control and Prevention (CDC) (Appendix C). This tool is a sexual health questionnaire also called the 5P's. This questionnaire is a simple approach to taking a sexual history. It helps to identify patient's level of risk, determine if screening is needed, and offer appropriate counseling to patients (CDC, 2023). This questionnaire was well developed and adapted from the CDC because the CDC applies rigorous scientific standards to ensure accuracy and reliability of research (CDC, 2017). Given the benefits of this questionnaire, it was utilized during the implementation of the fast-track STD screening program. The information from this questionnaire was utilized to draw inferences on patients who tested positive for chlamydia. It was also utilized to assess the sexual behaviors of participants. Overall, the surveys and questionnaire helped draw conclusions and may contribute to the development of future interventions to help decrease high risk sexual behaviors and chlamydia cases on campus.

Procedures

The first phase of the implementation process started with an STD educational outreach event. This consisted of two 30-minute sessions that included education on STDs and contraceptive resources. To assess the knowledge and attitudes towards STDs for this population, a pre- and post-STD Educational Survey was administered, and consent was obtained (see Appendix D for surveys). Pre-and Post STD educational surveys did not contain identifiable information as outlined in the consent statement. Students were made aware that only demographic data such as gender, age, and race would be utilized. It did not include any parts of

their names, dates of birth, or emails. Data was de-identified with a randomly assigned three-digit number on the pre-and post-STD educational survey. This three-digit number corresponded with a sign-in sheet for those participating. The medical assistant kept and stored the sign-in sheet in locked file cabinet. Students were selected at random if the inclusion criteria were met. Inclusion criteria included any actively enrolled students, including all genders, and between the ages of 18-25.

Along with education, resources were provided to students free of charge. This included items such as brochures on chlamydia, condoms, dental dams, and lubrication supplied by the Chicago Department of Public Health. The following steps occurred during the educational intervention January 23, 2024.

Step 1. Students arrived at the STD educational event conducted by (project leader) and were assigned a three-digit number to decode any identifying information.

Step 2. Students who opted to participate in the survey were required to sign a consent form, which was obtained and stored by a medical assistant in a locked file cabinet and could opt out at any time. The project leader discussed the purpose of the project and disclaimer of confidentiality.

Step 3. The Pre STD-Educational Survey was disbursed (Appendix D), coded with the same 3-digit code from the sign-in sheet without any personal identifiers to ensure confidentiality. No personal identifiers were used, and a disclaimer can be found at the top of the survey. Participation is voluntary.

Step 4. Two 30-minute informational sessions took place on the same day, which included a 15-minute PowerPoint presentation that provided education about chlamydia and covered key facts about the disease, signs and symptoms, treatment, and prevention methods. Students were then

allowed to ask any questions after the session.

Step 5. Distribution of resources such as educational brochures, contraceptives such as condoms, dental dams, and lubricants were offered to students.

Step 6. The Post-STD Educational Survey was distributed after the educational session and assigned to the same student with a corresponding 3-digit number from Pre-STD survey.

Step 7. After the event concluded, surveys were summarized utilizing paired t-tests to show a positive or negative change between pre-and post-testing.

The second implementation phase was introducing the fast-track STD screening program. This program was initiated on February 12, 2024. Since these were two separate phases in the project the participants may differ from those who attended the STD educational event, however, in some cases, students who participated in the STD educational event did seek fast-track STD screening services after educational intervention. Participants were selected at random if inclusion criteria were met. Inclusion criteria also included any actively enrolled students, inclusive of all genders, between the ages of 18-25 years old and required to be asymptomatic to participate in fast-track screening. Those students who were symptomatic did not meet the inclusion criteria to participate in the program and were scheduled for an appointment with the provider. Participants arrived for testing on a walk-in fast-track basis. Before screening, students signed a consent form (Appendix E) and then filled out a sexual history screening questionnaire tool (Appendix C), adapted from the Centers for Disease Control and Prevention (CDC). This tool was later utilized to draw inferences about the participants who tested positive for chlamydia. Any patients who tested positive for chlamydia or any other STD received a follow-up call that included treatment, education, and referral to behavioral health. The following steps occurred during the implementation of the fast-track screening program.

Step 1. The medical assistant was educated on the procedures before the initiation of testing. Discussed were the expectations, procedures, and a teach-back of steps of fast-track screening before the first screening participant.

Step 2. Participants arrived for testing on a walk-in fast track basis during normal business hours.

Step 3. When students arrived for testing, they were made aware that this was a quality improvement program, their participation was voluntary, and they could withdraw participation at any time.

Step 4. Students were given a consent form that included disclosures (Appendix E): once consent was signed, the appointment was continued.

Step 5. Consents were retrieved and stored by the medical assistant. All consents were electronically signed and kept in an encrypted electronic medical record.

Step 6. Students then filled out a Sexual Health Questionnaire adapted from the CDC, a disclaimer, and a consent form. (Appendix, C)

Step 7. The Sexual Health Questionnaire was distributed, collected by the medical assistant and reviewed by the project leader/ provider. A roster was maintained and stored in a locked file cabinet. Surveys were decoded with a randomly assigned three-digit code. The project leader kept the rosters in a locked file cabinet in the wellness center on the university campus. Responses from this questionnaire were analyzed to make a linkage between sexual behaviors and results from the visit.

Step 8. The appointment was fulfilled if the student was asymptomatic and met inclusion criteria.

Step 9. If students were symptomatic, they did not meet the criteria for participation and would be scheduled for an appointment with a healthcare provider/ project leader.

Step 10. The medical assistant collected urine samples for eligible participants and discarded the remainder of the specimen in the appropriate waste. Participants did not have any interactions with healthcare provider/ project leader. The certified medical assistant handled the entire appointment.

Step 11. Students were made aware that results would be analyzed by the project leader, and information would be strictly confidential.

Step 12. Once the visit was complete, students were instructed to wait 2-3 days for results.

Step 13. If the results are negative, no further action was needed.

Step 14. If the results were positive, students received a follow-up call and treatment from the healthcare provider/ project leader.

Step 15. Students found to be high risk were referred to the counseling department for behavioral health counseling.

Step 16. Once the sample size was met (31), the program concluded.

Step 17. A chart review was performed by the project leader with consent from participants, and data analyses were conducted to assess chlamydia case numbers only.

Step 18. The Sexual Health Questionnaire was analyzed to assess whether the positive cases had a linkage to reported high-risk sexual behaviors. Demographical data was also analyzed to identify the most vulnerable populations.

Resources/ Budget

Collaboration with the Chicago Department of Public Health (CDPH) was vital to the success of this project. CDPH was able to provide literature related to chlamydia and other STDs and contraceptive resources free of charge to help support the project. This included condoms, lubrication, and free HIV screening at-home kits. Since this is a university-based wellness center, all the services offered were free of charge to participants if they were actively enrolled. A set fee was charged to the university to their corporate account to help support the cost of these services. The cost includes the services provided along with the medical assistant and provider fees.

Results

Overview of Data Analysis

Data was imported into and analyzed using IBM SPSS version 23 for Windows (IBM Corp, 2015). Descriptive statistics and frequency tables were used to summarize the survey responses for 1) the pre-and post-sexually transmitted disease (STD) education survey and 2) the fast-track STD screening and sexual health questionnaire.

Normality of the data was examined using the Shapiro-Wilk test (Field, 2013). The non-parametric Wilcoxon Signed Rank Test technique was used to examine the pre-and post STD education survey to determine if there was a difference in attitudes towards sexual health before and after receiving the STD education. This technique is used when the sample size is small, the data collected are measured at the ordinal level and the scores are not normally distributed.

As the sample size was small, Fisher’s exact tests (Field, 2013) were used to determine if there was a relationship between chlamydia cases (as measured by the fast-track STD screening)

and sexual health (as measured by the individual survey item of the sexual health questionnaire). For any test, a p-value less than 0.05 indicated statistical significance.

Goal 1. Increase Knowledge of Chlamydia

Increase the knowledge of chlamydia and other STDs on campus through an STD educational awareness event. During this phase a pre- and post STD educational survey was utilized to assess whether an educational stimulus would change the individual's attitudes towards chlamydia and other STD's. The goal was to increase the knowledge of chlamydia and thereby decrease chlamydia cases on campus. There was a total of 31 participants during this educational event. Demographically African Americans accounted for 77.4% of participants, Caucasians accounted for 12.9% Asians accounted for 3.2% and pacific Islanders accounted for 6.5%. 71% of participants were female and 29% were male. When the survey responses for both the pre- and post STD educational survey were analyzed, it showed that there was statistically significant relationship with an STD educational intervention to help increase knowledge of chlamydia campus. Most students agreed that they felt more confident in their knowledge of chlamydia and other STDs after receiving more education post intervention. Nine (29.0%) of the thirty-one students who sought testing attended the STD educational intervention. Of the nine students who received the educational intervention only one student One (3.23%), student test positive for chlamydia. This supports the project goal of increasing knowledge of chlamydia on campus to help reduce the cases of chlamydia on campus.

Demographics

31 subjects participated in the pre-and post STD education survey. Table 6 presents their demographics and Figures 5-8 provide the visualization of the information in Table 6. The average age of the participants was 20.06 years (*SD* = 1.48), with the youngest being 18 years

old and the oldest being 23 years old. Majority of the participants were female (71.0%), Black (77.4%), and not Hispanic or Latino (87.1%). Over 40% of the participants (41.9%) were in their freshman year.

Table 6

Demographics

	N	%
Age (*M* ± *SD*)	20.06 ± 1.48	
Gender		
Female	22	71.0
Male	9	29.0
Race		
Black	24	77.4
Asian	1	3.2
Pacific Islander	2	6.5
Caucasian	4	12.9
Hispanic or Latino		
Yes	4	12.9
No	27	87.1
Year in college		
Freshman	13	41.9
Sophomore	10	32.3
Junior	7	22.6
Senior	1	3.2

Note. Table describes the demographical data of participants from the STD educational intervention. n= the number of participants. It also displays the participants year in college at the time of intervention.

Figure 5

Pre-and Post STD Educational Survey Gender Results

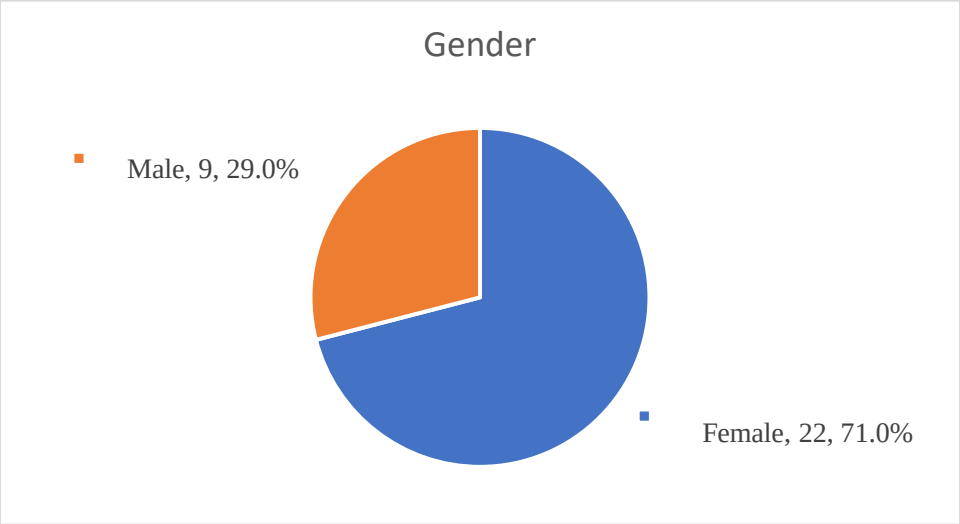


Figure 6

Pre-and Post STD Educational Survey Race Results

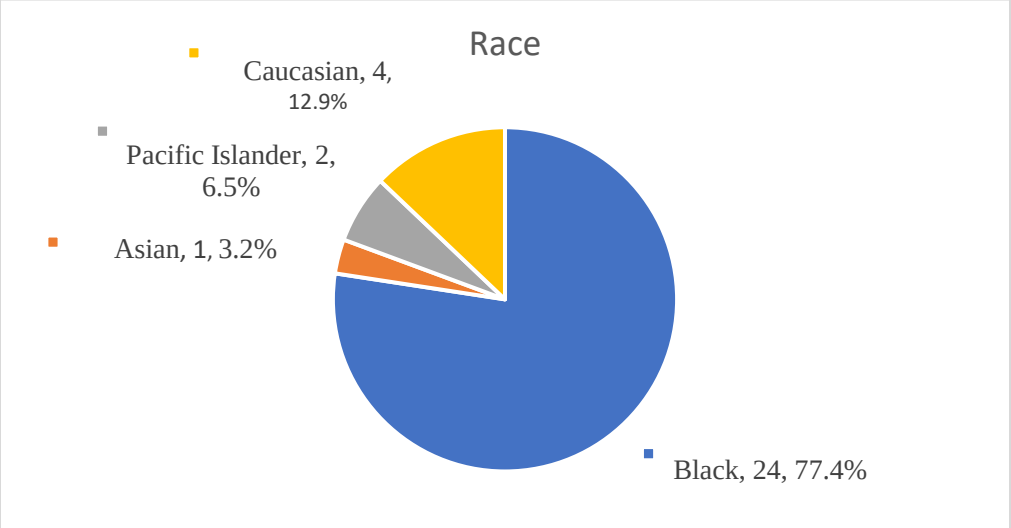


Figure 7

Pre-and Post STD Educational Survey Hispanic or Latino Origin Results

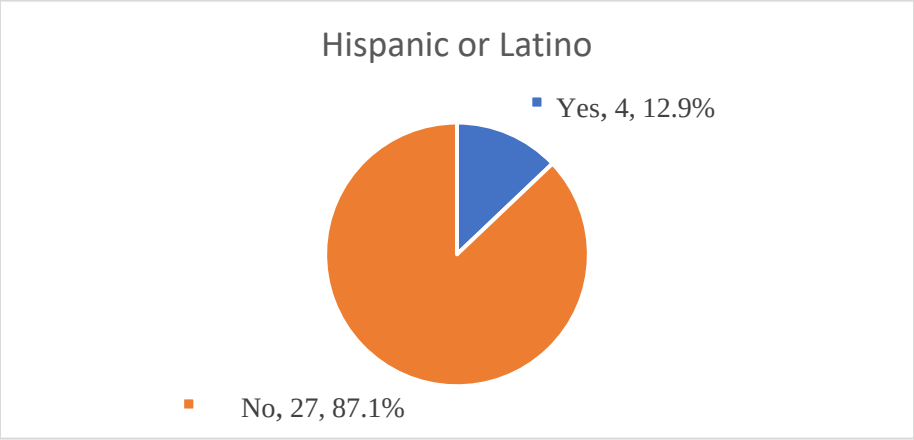
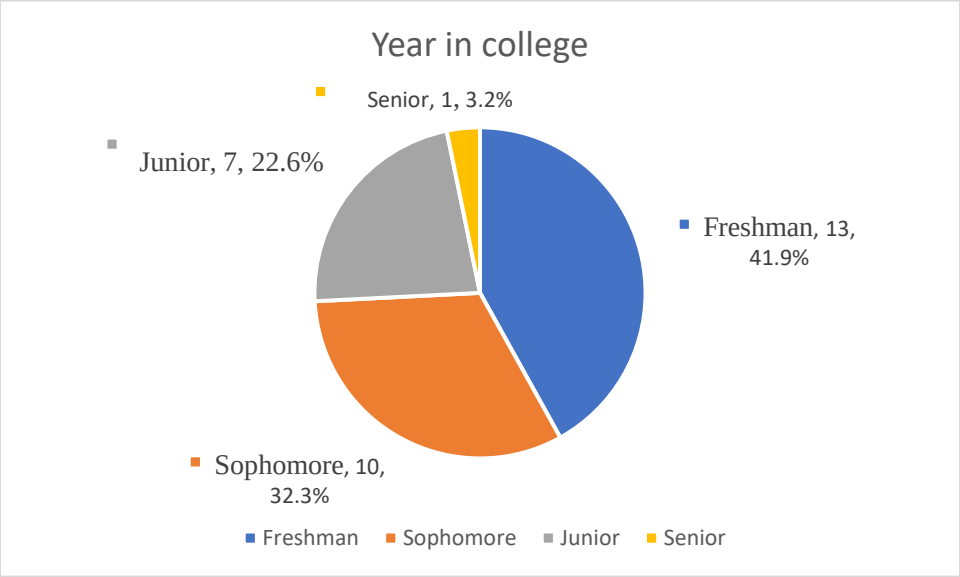


Figure 8

Pre-and Post STD Educational Survey Year in College results



Participants' attitudes towards sexual health were summarized in Table 7 and visualized in Figure 9. Before the STD education, participants were neutral about the statement "I always use condoms" (Q1, M = 3.19, Table 7). However, after the STD education, participants moderately agreed about the statement "I always use condoms" (Q1, M = 3.71, Table 7). The results of the Wilcoxon signed rank test indicated that the attitude difference regarding the

statement “I always use condoms” was statistically significant (*W* = 41.000, *SE* = 8.261, *z* = 2.239, *p* = 0.025; Table 8).

Participants moderately agreed about the statement “I have been tested for sexually transmitted diseases” before (*M* = 3.81) and after (*M* = 4.03) the STD education (Q2, Table 7). Although after the STD education, participants agreed more about the statement “I have been tested for sexually transmitted diseases” than before the STD education, the difference was not statistically significant (*W* = 17.000, *SE* = 4.704, *z* = 1.382, *p* = 0.167; Table 8).

Participants moderately agreed about the statement “I encourage my partner to be tested” before (*M* = 3.97) and after (*M* = 4.19) the STD education (Q3, Table 7). Although after the STD education, participants agreed more about the statement “I encourage my partner to be tested” than before the STD education, the difference was not statistically significant (*W* = 46.500, *SE* = 10.770, *z* = 0.975, *p* = 0.330; Table 8).

Participants moderately agreed about the statement “All sexually transmitted diseases are preventable” before (*M* = 3.87) and after (*M* = 4.13) the STD education (Q4, Table 7). Although after the STD education, participants agreed more about the statement “All sexually transmitted diseases are preventable” than before the STD education, the difference was not statistically significant (*W* = 92.500, *SE* = 20.448, *z* = 0.782, *p* = 0.434; Table 8).

Participants strongly agreed about the statement “I am confident in my knowledge on sexually transmitted diseases” before (*M* = 4.29) and after (*M* = 4.45) the STD education (Q5, Table 7). Although after the STD education, participants agreed more about the statement “I am confident in my knowledge on sexually transmitted diseases” than before the STD education, the difference was not statistically significant (*W* = 106.000, *SE* = 22.113, *z* = 0.927, *p* = 0.354; Table 8).

After the STD education, participants moderately agreed about the statement "I will always use condoms with sexual intercourse" (*M* = 4.06) (Q6, Table 7).

Table 7

Attitudes Towards Sexual Health

	Frequency (%) of survey responses					
	1	2	3	4	5	*M* (*SD*)
Pre						
Q1	3 (9.7)	8 (25.8)	7 (22.6)	6 (19.4)	7 (22.6)	3.19 (1.33)
Q2	3 (9.7)	4 (12.9)	2 (6.5)	9 (29.0)	13 (41.9)	3.81 (1.38)
Q3	1 (3.2)	3 (9.7)	3 (9.7)	13 (41.9)	11 (35.5)	3.97 (1.08)
Q4	1 (3.2)	5 (16.1)	6 (19.4)	4 (12.9)	15 (48.4)	3.87 (1.28)
Q5	0	3 (9.7)	5 (16.1)	3 (9.7)	20 (64.5)	4.29 (1.07)
Post						
Q1	2 (6.5)	4 (12.9)	6 (19.4)	8 (25.8)	11 (35.5)	3.71 (1.27)
Q2	2 (6.5)	2 (6.5)	2 (6.5)	12 (38.7)	13 (41.9)	4.03 (1.17)
Q3	0	2 (6.5)	0	19 (61.3)	10 (32.3)	4.19 (0.75)
Q4	0	2 (6.5)	3 (9.7)	15 (48.4)	11 (35.5)	4.13 (0.85)
Q5	0	0	0	17 (54.8)	14 (45.2)	4.45 (0.51)
Q6	0	1 (3.2)	5 (16.1)	16 (51.6)	9 (29.0)	4.06 (0.77)

Note. Q1: I always use condoms; Q2: I have been tested for sexually transmitted diseases. Q3: I encourage my partner to be tested; Q4: All sexually transmitted diseases are preventable; Q5: I am confident in my knowledge on sexually transmitted diseases. Q6: I will always use condoms with sexual intercourse. For survey responses, 1 = strongly disagree, 2 = disagree, 3 = neither disagree nor agree, 4 = strongly agree, and 5 = agree. *M* = mean; *SD* = standard deviation.

Table 8

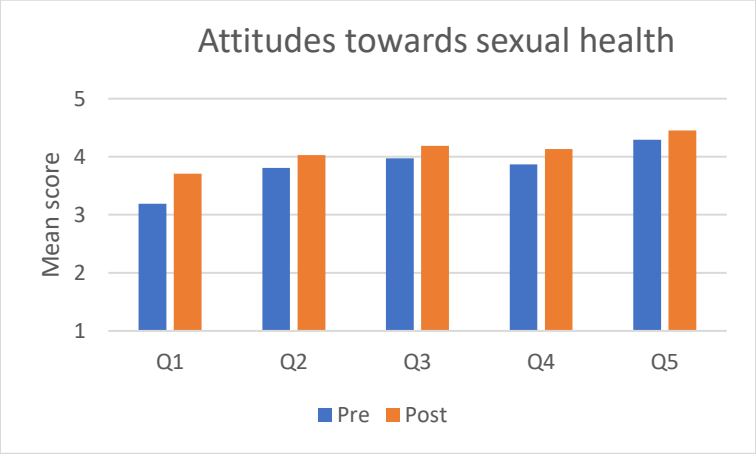
Comparisons of Attitudes Towards Sexual Health Before and After Receiving the STD Education

			Shapiro-Wilk test			Wilcoxon signed rank test				
	M_{diff}	SD_{diff}	*S*	*df*	*P*	*W*	*SE*	*z*	*p*	*r*
Q1	0.52	1.21	0.638	31	< 0.001	41.000	8.261	2.239	0.025	0.28
Q2	0.23	0.92	0.530	31	< 0.001	17.000	4.704	1.382	0.167	0.18
Q3	0.23	1.09	0.700	31	< 0.001	43.500	10.770	0.975	0.330	0.12
Q4	0.26	1.41	0.836	31	< 0.001	92.500	20.448	0.782	0.434	0.10
Q5	0.16	1.16	0.821	31	< 0.001	106.000	22.113	0.927	0.354	0.12

Note. Q1: I always use condoms; Q2: I have been tested for sexually transmitted diseases. Q3: I encourage my partner to be tested; Q4: All sexually transmitted diseases are preventable. Q5: I am confident in my knowledge on sexually transmitted diseases. M_{diff} = mean difference; SD_{diff} = standard deviation of the differences; S = Shapiro-Wilk test statistic; df = degrees of freedom; p = p-value; W = Wilcoxon signed rank test statistic; SE = standard error; z = z-statistic; p = p-value; r = effect size, computed as $r = \frac{z}{n}$, where n = the number of observations over the two time points (i.e., n = 62 (= 31+31) observations for pre-test and post-test), not the number of cases (i.e., 31) (Pallant, 2007). Using Cohen (1988) criteria for r, 0.1 = small effect, 0.3 = medium effect, and 0.5 = large effect. Based on the results of Shapiro-Wilk tests, the data for the differences in the response scores before and after the STD education were not normally distributed ($p < 0.05$).

Figure 9

Mean Scores for Attitudes Towards Sexual Health Before and After Receiving the STD Education



Goal 2*: *Increasing Access to Healthcare

Goal two involved introducing a fast-track STD screening program on campus to increase access to care and help decrease chlamydia cases on campus. During this fast- track screening intervention a sexual health questionnaire was introduced to help elicit questions about the sexual health of participants. This questionnaire was used as a guide to help identify potential high risk sexual behaviors and help identify ways to prevent chlamydia on campus.

Demographics

31 subjects participated in the Fast Track STD screening and sexual health questionnaire. Table 9 presents their demographics and Figures 10-13 provide the visualization of the information in Table 9. The average age of the participants was 20.74 years (*SD* = 2.03), with the youngest being 18 years old and the oldest being 25 years old. Majority of the participants were female (64.5%), Black (93.5%), and not Hispanic or Latino (96.8%). Over 40% of the participants (41.9%) were in their freshman year.

Table 9

Demographics

	N	%
Age (*M* ± *SD*)	20.74 ± 2.03	
Gender		
Female	20	64.5
Male	11	35.5
Race		
Black	29	93.5
Asian	0	0
Pacific Islander	1	3.2
Caucasian	1	3.2
Hispanic or Latino		
Yes	1	3.2
No	30	96.8
Year in college		
Freshman	13	41.9
Sophomore	5	16.1
Junior	9	29.0
Senior	4	12.9

Note. This table displays the demographical data and year in college of participants from the Fast-track STD screening. N=the total number of participants.

Figure 10

Gender - Fast Track STD Screening

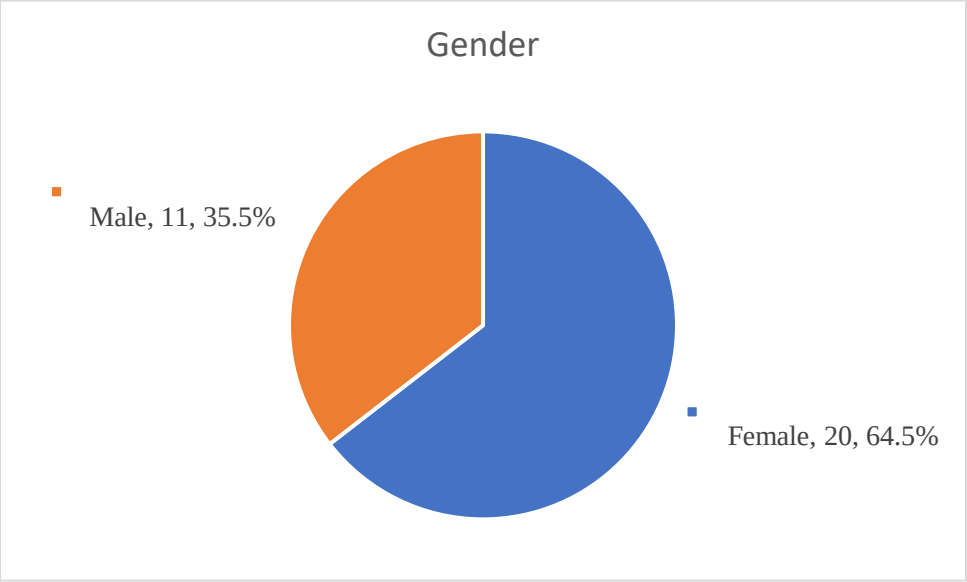


Figure 11

Race - Fast Track STD Screening

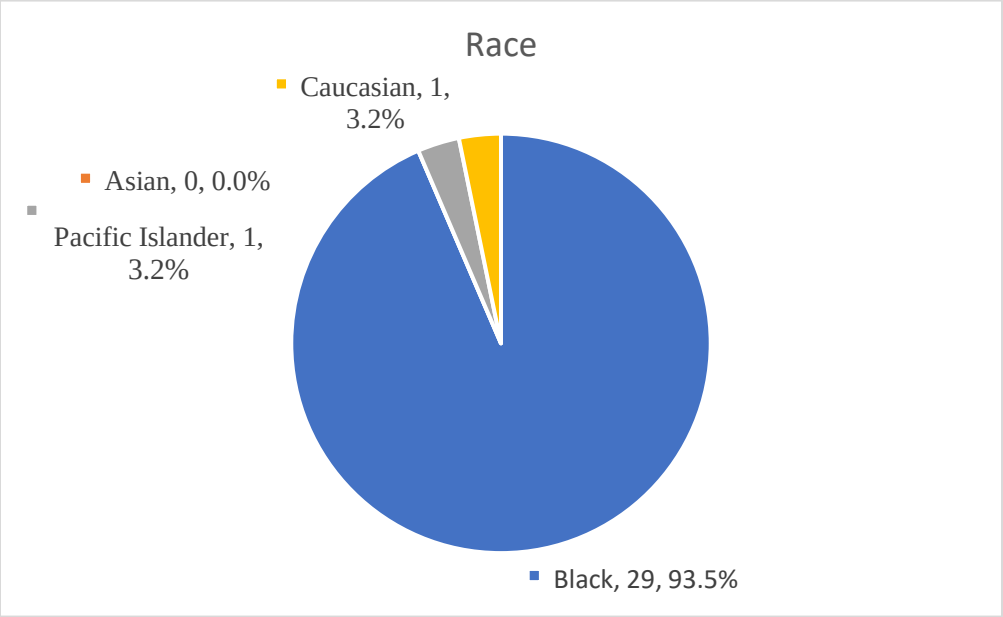


Figure 12

Hispanic or Latino - Fast Track STD Screening

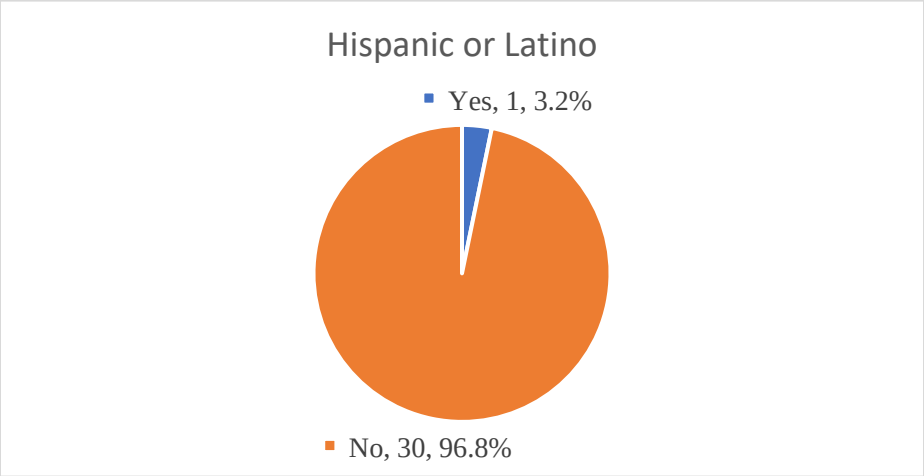


Figure 13

Year in College - Fast Track STD Screening

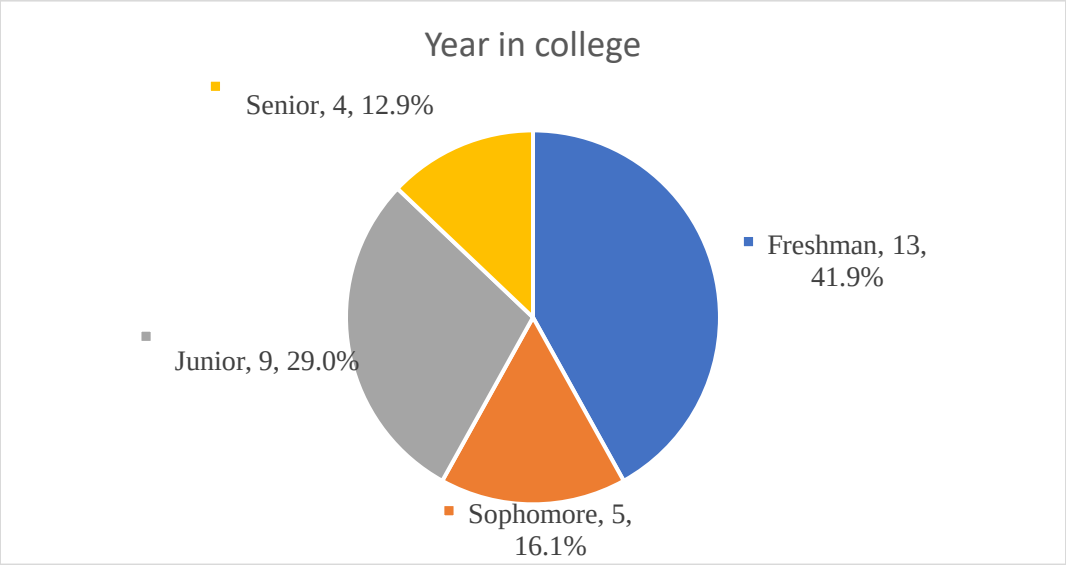


Table 10 shows the Fast Track STD screening results. Nearly 20% of the participants were Chlamydia positive (19.4%), had been followed up (19.4%), were referred for further care (19.4%), and were receiving treatment (19.4%). Nearly one-third of the participants (29.0%) attended education. Few participants (3.2%) had co-infection.

Table 10

Fast Track STD Screening Results

	N	%
Results of Chlamydia		
Positive	6	19.4
Negative	25	80.6
Follow-up		
Yes	6	19.4
No	25	80.6
Referral		
Yes	6	19.4
No	25	80.6
Attended education		
Yes	9	29.0
No	22	71.0
Co-infection		
Yes	1	3.2
No	30	96.8
Treatment		
Yes	6	19.4
No	25	80.6

Note. This table displays the fast-rack STD screening results of participants. N=the total number of participants.

The results of sexual health questionnaire are summarized in Table 11. Most of the participants (90.3%, Q1) currently had sex of any kind (i.e., oral, vaginal, or anal) with anyone. Over half of the participants had 1 sex partner in the last couple months (54.8%, Q2) and had male as their sex partners (51.6%, Q3). Nearly one-third of the participants or their partners (32.3%, Q4) had other partners. Majority of the participants had vaginal sex (96.8%, Q5), and among them, 16.7% always used condoms (Q6). Less than 20% of participants had anal sex (16.1%, Q7), and among them, 20.0% always used condoms (Q8). Majority of the participants had oral sex (93.5%, Q9).

Nearly two-thirds of the participants and their partner (64.5%, Q10) discussed STI prevention. Majority of the participants had received HPV, Hepatitis A, or Hepatitis B vaccines (93.5%, Q11), were not familiar with Prep, a medication that can prevent HIV (80.6%, Q12), and had been tested for STIs (93.5%, Q13). All participants would like to be tested (100%, Q14). Over half of the participants (54.8%, Q15) had been diagnosed with a STI in the past, and among them, 82.3% got treatment (Q16).

No one had any symptoms that keep coming back (100%, Q17). Less than 10% of the participants (6.5%, Q18) had their current partner test positive for any STIs. Nearly 60% of the participants (58.1%, Q19) knew their partners HIV status. Nearly 70% of the participants (67.7%, Q20) thought that they want to have children at some point and about 40% were undecided about when that would happen (41.9%, Q21). Less than 40% of the participants (38.7%, Q22) were using contraception or practicing any form of birth control. Only 1 participant (3.2%, Q23) needed information on birth control.

Fisher's exact tests were used to determine if there was a relationship between chlamydia cases and sexual health and the results are also presented in Table 11. There was no statistically significant relationship between chlamydia cases and sexual health (as measured by the individual survey item of the sexual health questionnaire) ($p > 0.05$).

Table 11

Sexual Health Questionnaire

		Screening results		
	All samples (N = 31)	Positive (N = 6)	Negative (N = 25)	p
Q1. Are you currently having sex of any kind- oral, vaginal, or anal with anyone?				
Yes	28 (90.3)	6 (100)	22 (88.0)	1.000
No	3 (9.7)	0	3 (12.0)	

Table 11 (continued)

Q2. In the last couple months how many sex partners have you had?				
0	2 (6.5)	0	2 (8.0)	0.364
1	17 (54.8)	2 (33.3)	15 (60.0)	
2	10 (32.3)	3 (50.0)	7 (28.0)	
≥ 3	2 (6.5)	1 (16.7)	1 (4.0)	
Q3. What is/are the gender(s) of your sex partners?				
Male	16 (51.6)	4 (66.7)	12 (48.0)	0.411
Female	12 (38.7)	1 (16.7)	11 (44.0)	
Both	3 (9.7)	1 (16.7)	2 (8.0)	
Q4. Do you or your partner(s) currently have other partners?				
Yes	10 (32.3)	4 (66.7)	6 (24.0)	0.067
No	21 (67.7)	2 (33.3)	19 (76.0)	
Q5. Have you had vaginal sex?				
Yes	30 (96.8)	6 (100)	24 (96.0)	1.000
No	1 (3.2)	0	1 (4.0)	
Q6. If yes, do you use condoms? (N = 30)				
Never	7 (23.3)			
Sometimes	18 (60.0)			
Always	5 (16.7)			
Q7. Have you had anal sex?				
Yes	5 (16.1)	2 (33.3)	3 (12.0)	0.241
No	26 (83.9)	4 (66.7)	22 (88.0)	
Q8. If yes, do you use condoms? (N = 5)				
Never	1 (20.0)			
Sometimes	3 (60.0)			
Always	1 (20.0)			
Q9. Have you ever had oral sex?				
Yes	29 (93.5)	6 (100)	23 (92.0)	1.000
No	2 (6.5)	0	2 (8.0)	
Q10. Do you and your partner discuss STI prevention?				
Yes	20 (64.5)	5 (83.3)	15 (60.0)	0.383
No	10 (35.5)	1 (16.7)	10 (40.0)	
Q11. Have you received HPV, Hepatitis A, or Hepatitis B vaccines?				
Yes	29 (93.5)	6 (100)	23 (92.0)	1.000
No	2 (6.5)	0	2 (8.0)	

Table 11 (continued)

Q12. Are you familiar with Prep, a medication that can prevent HIV?				
Yes	6 (19.4)	1 (16.7)	5 (20.0)	1.000
No	25 (80.6)	5 (83.3)	20 (80.0)	
Q13. Have you ever been tested for STIs?				
Yes	29 (93.5)	6 (100)	23 (92.0)	1.000
No	2 (6.5)	0	2 (8.0)	
Q14. Would you like to be tested?				
Yes	31 (100)	6 (100)	25 (100)	N/A
Q15. Have you ever been diagnosed with a STI in the past?				
Yes	17 (54.8)	5 (83.3)	12 (48.0)	0.185
No	14 (45.2)	1 (16.7)	13 (52.0)	
Q16. If yes, did you get treatment? (N = 17)				
Yes	14 (82.3)			
No	2 (11.8)			
N/A	1 (5.9)			
Q17. Have you had any symptoms that keep coming back?				
No	31 (100)	6 (100)	25 (100)	N/A
Q18. Has your current partner tested positive for any STIs?				
Yes	2 (6.5)	1 (16.7)	1 (4.0)	0.355
No	29 (93.5)	5 (83.3)	24 (96.0)	
Q19. Do you know your partners HIV status?				
Yes	18 (58.1)	4 (66.7)	14 (56.0)	1.000
No	13 (41.9)	2 (33.3)	11 (44.0)	
Q20. Do you think that you want to have children at some point?				
Yes	21 (67.7)	5 (83.3)	16 (64.0)	0.634
No	10 (32.3)	1 (16.7)	9 (36.0)	
Q21. When do you think that might be?				
Never	9 (29.0)	1 (16.7)	8 (32.0)	0.863
> 3 years	9 (29.0)	2 (33.3)	7 (28.0)	
Undecided	13 (41.9)	3 (50.0)	10 (40.0)	
Q22. Are you and your partner using contraception or practicing any form of birth control?				
Yes	12 (38.7)	3 (50.0)	9 (36.0)	0.653
No	19 (61.3)	3 (50.0)	16 (64.0)	
Q23. Do you need any information on birth control?				
Yes	1 (3.2)	0	1 (4.0)	1.000

Table 11 (continued)

No	30 (96.8)	6 (100)	24 (96.0)

Note. N/A = not available. Fisher's exact tests were not applied to sub-questions (e.g., Q6, Q8, and Q16).

Goal 3. Increase Access to Mental Health Services

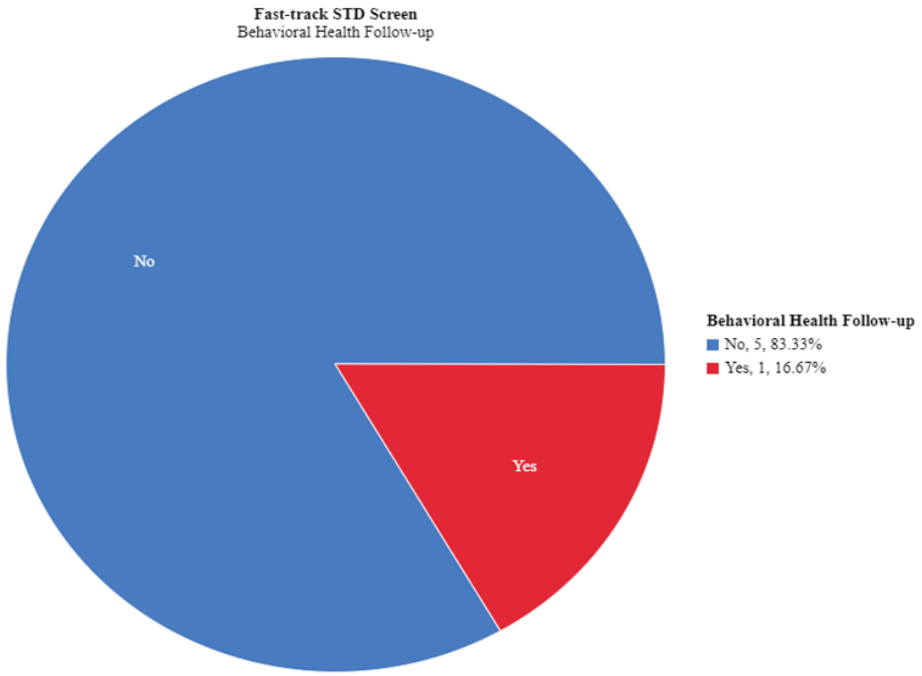
The final goal was to increase access to mental health services through the counseling department for those considered high-risk. Of the 6 individuals who tested positive for chlamydia and were referred, only one sought behavioral health services. This shows that the goal to increase access to services was not met because individuals did not seek care after referrals were made. This shows a gap in care and more effort needs to be put into increasing collaboration with behavioral health specialists to help identify high risk individuals to help decrease chlamydia cases on campus.

Demographics

The demographical data from the goal 2 and goal 3 are the same. It is important to highlight that African Americans accounted for 100% of the positive chlamydia cases. This supports current research that minority groups are the most impacted by STDs. The linkage between the STD educational event and fast-track STD screening was observed from the participants who attended education and decided to seek testing after the educational intervention. Of the 31 participants 9 of them participated in the STD educational event. This shows that increasing awareness of chlamydia was an appropriate stimulus to urge students to get tested. Figure 14 displays the percentage of individuals who test positive and followed up for behavioral health services.

Figure 14

Positive Chlamydia Participants that Attended Behavioral Health Counseling



Formative Evaluation

Formative evaluation occurs during the development of the project, whereas summative evaluation occurs when the project has concluded (University of Michigan, 2021). The formative evaluation plan for this project included looking at any current STD programs and screening to assess if any changes needed to be made. The idea of developing and implementing a new screening program was proposed. The goal is to implement a new screening program to increase access to care. Given this is a predominantly black college, most students are African Americans. Disparities among African Americans have been proven to exist. These include access to care, underlying health disparities, poverty, and discrimination (Kaiser Family Foundation, 2022). Investigating the current screening processes and resources that are in place currently helped develop interventions that need change. To help support the objective of increasing knowledge of

STDs on campus to reduce chlamydia cases on campus, an educational outreach event was planned. To evaluate the effectiveness of the intervention, a Pre STD educational survey and a Post STD educational survey were conducted to assess the attitudes of this population towards STIs before and after educational intervention. These pre- and post-surveys can be found in (Appendix D).

For the duration of the project implementation, weekly meetings were held with the capstone committee member, APRN. This included discussions about the project, its progression, any barriers faced, and ways to improve. During the fast-track STD screening, weekly meetings were held with the medical assistant, to assess the progress of the fast-track program. Barriers were discussed such as a decrease in participants due to spring break. During our weekly discussion, we recognized that communication of the program introduction on campus may have also been an issue. It was important to consider that students may have not received the email invitation, or it was overlooked. To help increase the number of participants, the email invitation was printed, and hand-delivered to students on campus to make them aware of the program on campus. Doing this was pivotal to the success of meeting the sample. After informing students of the fast-track screening program, more students started to seek testing.

Summative Evaluation

Summative evaluation occurs at the end of a program. The goal of the summative plan was to improve outcomes. To evaluate the outcomes of this project, a Plan-Do-Study-Act (PDSA) was completed. The PDSA is a method to test a change that is implemented. It guides the thinking process and then evaluates the outcomes and ways to improve (AHRQ, 2024). The project planned and introduced a fast-track STD screening program on campus to help increase

the uptake of testing and increase access to care. The project also aimed to see if increasing the knowledge of STDs on campus would change the attitudes of young adults and decrease chlamydia on campus. The hope was that STD education would be the stimulus needed to decrease unhealthy sexual behaviors on campus. To execute this an educational intervention was introduced and students participated in a pre-and post-STD educational survey.

What was observed during the presentation was that students lacked the basic knowledge of STDs. What was also observed from data analysis was that after students learned about the different STDs, they felt more knowledgeable and equipped to avoid STDs. The measurement goal was met because more than half of participants shifted their attitudes from not wearing condoms, to wanting to wear condoms. Post-STD educational intervention all the participants felt more confident in their knowledge of STDs. What I can conclude from this project is that more educational interventions need to be in place on the college campus. Adapting to the fast-track STD screening as well as increasing STD educational interventions would be beneficial for the university campus. Seeing a slight reduction in chlamydia cases supports that the interventions that were implemented can have a positive impact on the campus community.

Ways to improve this program lie in the way information is delivered to students. Using social media platforms as well as hand delivering information could have improved participation. Most students did not come in after the initial invitation that was emailed to them. Another way to improve this program is to help decrease the stigma related to STDs. Many students may not seek care because of fear of embarrassment. Implementing new strategies to decrease stigma and promote healthy sexual behaviors is needed. Overall, the combination of increasing access to care, increasing STD knowledge, and increasing access to mental health services has shown a

positive shift, and the continuation of this program on campus may be beneficial. Over time, the goal is to decrease the overall cases of chlamydia.

Discussion of Findings

Findings from this project showed that the interventions implemented to help reduce chlamydia on campus were beneficial. The goals were to decrease chlamydia cases by increasing access to care, increasing knowledge of STDs, and increasing mental health services. The goal to increase knowledge of STDs on campus was met because data analysis supports that there was a positive shift in students' attitudes once education stimulus was introduced. All participants endorsed that they felt more confident in their knowledge of STDs after the informational session. Data also supports that the educational stimulus triggered individuals to seek testing.

Data also supports that there was a decrease in chlamydia cases following the implementation of the Fast-track STD screening program and STD educational intervention. Improving access to screening helped to increase the uptake of testing on campus. In comparison to the fall semester of 2023 and the spring semester of 2024, there was a slight decline in the chlamydia cases on campus. For the fall academic school year, chlamydia cases on campus were at 32%, while the spring chlamydia case rate was 20%. This shows a 12% reduction rate in chlamydia cases. One can draw inferences that increasing access to care and increasing awareness had an impact on the reduction of chlamydia cases on campus. The HBM that guided this project worked for those who did attend the STD educational intervention as well as those who did not. The stimulus for participants needed to trigger STD screening was education of STDs. The stimulus needed to trigger participants who did not attend the STD screening was the increase and ease of access to care.

The third goal to increase access to mental health services faced barriers. The barrier faced during this implementation phase was the stigmatization attached to seeking mental health services. Poor follow-up with behavioral health from students who tested positive for chlamydia or were high risk was evident. Although access to mental health services was available, only one of the participants followed up. Research has shown that stigma still exists as it relates to mental health. More efforts need to be put into helping individuals feel more confident to seek mental health services. Demographically chlamydia continues to impact minority groups. More research is needed to identify effective interventions to help decrease this disparity. Overall, the findings of this project show that increasing access to care and increasing knowledge of STDs can help decrease chlamydia cases on a university campus. In terms of mental health findings show that more effort needs to be put into helping students feel more comfortable seeking mental health services.

Limitations

There are limitations to this project. It must consider the resistance of students, and the stigma related to STDs. Limitations are more closely related to those who did not utilize the fast-track screening program as well as those who did not attend education. If individuals are not equipped with the necessary knowledge of chlamydia and STDs, they won’t perceive it as a threat and won’t take the necessary action to combat this disease. To help with limitations education will always be at the forefront. Another limitation faced was decrease uptake of testing due to spring break. Less students utilized services at this time which negatively impacted the sample. Chlamydia has an overall negative impact on the community, especially young adults. Developing an action plan that bridged the gaps between access to care, lack of knowledge, and resources was a key component to the success of this project.

Implications for Practice

Implications in practice were considered while implementing this project. The results of this project impacted the way we screen and deliver information to college students regarding sexually transmitted infections. This STD screening program utilized evidence-based practice to address the gaps in current practice. This project gave a new insight into how to address the sexual health needs of college students, and how to bridge the gap for this population. For this patient population, it was important to address this issue because it has many consequences if left untreated. The implications of not following up with patients who test positive can also lead to negative outcomes. The cost of providing this service will not change for the student as all services offered at the university are free of charge. This project has the potential to extend to other university clinic-based wellness centers and change the way we screen this high-risk population.

Dissemination/ Project Deliverable

The target journal for submission of this capstone project is the Journal of American College Health. One of the project deliverables is a publishable article related to this capstone project. This journal was selected because its journal seeks to publish valuable clinical lessons for college health professionals (American College Health Association, 2022). This journal supports submissions on preventative medicine, health promotion, student behaviors, and the mental health of college students (ACHA, 2022). Another project deliverable associated with this project is an orientation manual for the fast-track STD program. This manual serves as a guide to other providers so that they can continue this program in the future. This manual includes a STD

educational PowerPoint, and necessary screening tools. It also contains the roles and responsibilities, the purpose, the goals, and the steps to successfully implement this program.

Conclusion

Overall, implementing the fast-track STD screening program showed positive outcomes on the university campus. This project was successful at increasing access to care through fast-track screening, increasing knowledge of STDs through educational interventions, and increasing access to mental health services. The objective to decrease chlamydia cases was achieved which can be seen through data analysis. Many students felt more comfortable after being educated about chlamydia and other STDS. More time is needed to support the true success of this program. More research is needed to support interventions to combat this clinical issue. I would like to see more policies in place to support the fast-track STD screening program. The current USPSTF guidelines were supported because most participants who tested positive were female and African American. Continuing to offer sexually active women 24 years and younger is key to decreasing chlamydia on campus. Strategies to improve the uptake of mental health services are needed. Investigation into the stigma related to mental health is needed to help gain a better understanding and approach to getting students to seek services. This screening program should continue because data supports that it can help reduce chlamydia cases on the university campus. Reducing chlamydia cases can lead to a healthier campus community.

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Appendix A

Electronic Databases Searched

Date Searched	Database		Data Type	Key Search Terms	Results
10/18/2022	PubMed		Articles/ Academic Journals	"Chlamydia Prevention College Students"	34
10/18/2022	PubMed		Articles/ Academic Journals	"Evidence based practice chlamydia prevention in college students"	1
10/22/2022	CINAHL		Articles/ Academic Journals	"Chlamydia AND College Students"	69
10/22/2022	CINAHL		Articles/ Academic Journals	"Chlamydia AND University"	14,438
11/5/2022	CINAHL		Articles/ Academic Journals	"Chlamydia Interventions for College Students"	0
11/6/2022	CINAHL		Articles/ Academic Journals	"Chlamydia AND Intervention in College Age Students"	0
11/6/2022	Science Direct		Journals & Books	"Chlamydia Prevention in College Students"	1,446
11/9/2022	PubMed		Articles/ Academic Journals	"Patient delivered partner therapy college"	274

Appendix A

Summary of Studies

Author/Year	Study Purpose	Research Question/Hypothesis	Variables Studied	Sample Size & Age	Theoretical Framework	Setting
Habel et al, 2015	To evaluate the effect of social marketing campaign Get yourself Tested (GYT) designed to raise awareness and encourage testing on a college campus	Are their associations between GYT campaign awareness, and campaign events and STI test seeking	School level variables and STI numbers and positivity	1386 college students Ages Young adults ages 25 years and younger	Developed using constructs from the Health Belief Model and Theory of Planned Behavior	12 college campuses
Design	**Research Instrument**	**Instrument Psychometrics**	**Procedure for Data Collection**	**Study Results**	**Level of Evidence**	**Grade Recommendation**
Pilot Study	Self-Report Questionnaire	Simple frequency counts were performed. Using the self-report chi-squared test, odd ratios and bivariate correlations were used to test associations between GYT campaign awareness and hosting a GYT event as a school level predictor.	Data collection uncontrolled settings at different institutions Survey data were collected via paper and pencil questionnaires or personal digital assistant	Students who presented for STI testing were more likely to have heard of GYT than students who did not. Associations between GYT campaigns and STI seeking showed a positive correlation in	Level 3	Grade B Unclear if there was a benefit

Summary of studies (continued)

				students who sought testing		
Author/ Year	**Study Purpose**	**Research Question/Hypothesis**	**Variables Studied**	**Sample Size & Age**	**Theoretical Framework**	**Setting**
Jaworski et al, 2001	To evaluate a brief theoretically guided STI risk reduction intervention for college age women	Can Informational-Motivation-behavioral (IBM) skills model intervention can lead to reductions in risky sexual behaviors	Undergraduate female psychology students	N=78 undergraduate females Mean age of 20 years old	None	University of Syracuse campus, women taking psychology courses
Design	**Research Instrument**	**Instrument Psychometrics**	**Procedure for Data Collection**	**Study Results**	**Level of Evidence**	**Grade Recommendation**
Descriptive	Pre-and Post intervention survey.	Participants assigned to three groups. One session interventional information-based model (IBM) 150-minute one week after pre intervention survey A one session information only intervention (INFO) 150-minute intervention A wait list Control Group (WLC)	Participants received a post survey immediately after intervention. 2 month follow up assessment	The IMB and INFO groups increases STI knowledge amongst the study group. At the 2-month follow up knowledge about STI sustained and the IMB group showed reductions in number of sexual partners. The results provide support that an IMB based model can lead to reduction in risky sexual	Level 2	Grade A

Summary of studies (continued)

				behaviors and thereby decrease the incidence of STIs.		
Author/ Year	**Study Purpose**	**Research Question/Hypothesis**	**Variables Studied**	**Sample Size & Age**	**Theoretical Framework**	**Setting**
Hennrikus et al, 2009	To assess the benefit of chlamydia screening at preparticipation examinations for athletes in a college setting	Whether it was beneficial to educate and screen student athletes for Chlamydia during pre-participation examination or to not screen to help with reduction in chlamydia cases	Demographical and behavioral variables studied	N=439 220 males 219 women Ages 18-23 years old	None provided	University setting in Fresno California
Design	**Research Instrument**	**Instrument Psychometrics**	**Procedure for Data Collection**	**Study Results**	**Level of Evidence**	**Grade Recommendation**
Descriptive	Questionnaire	Descriptive statistics and chlamydia prevalence of positive chlamydia results amongst student athletes	Demographic and behavioral data questionnaire. Questions included ethnicity, age, sexual activity, STI symptoms, and STI history	Of the 439 students tested 13 test positive for Chlamydia. Chlamydial screening was of benefit to prevent the transmission of chlamydia and the continual rise of chlamydia cases	Level 1	Grade A
Author/ Year	**Study Purpose**	**Research Question/ Hypothesis**	**Variables Studied**	**Sample Size & Age**	**Theoretical Framework**	**Setting**

Summary of studies (continued)

Folasayo et al. 2017	To assess the knowledge, attitudes, risky behaviors, and preventive practices related to sexually transmitted diseases (STD) among health and non-health science university students	Not clearly stated but implied that should education curriculum in both schools and universities incorporate STD education so appropriate STD interventions are implemented	Source, knowledge, and attitudes about STDs	N=700 Ages 17-30 years old	None provided	Private University in Malaysia
Design	**Research Instrument**	**Instrument Psychometrics**	**Procedures for Data Collection**	**Study Results**	**Level of Evidence**	**Grade Recommendation**
Cross-Sectional Study Design Simple random Sampling	Self-Administered Questionnaire	Data was analyzed using a SPSS version 21 Statistical Software program. Chi-square was used to analyze the association between variables	Students were given a 20-item attitude related questionnaire regarding sexual behaviors	More than half the students did not know Gonorrhea was a STD and 26% of the students did not know a lot about Chlamydia. A gap in knowledge exist in college students and a critical need of sexual education programs is needed	Level 4	Grade B Evidence not high quality

Summary of studies (continued)

Author/ Year	Study Purpose	Research Question/Hypothesis	Variables Studied	Sample Size & Age	Theoretical Framework	Setting
Reeves JM et al, 2023	Reviewing current and past literature	Do STD/HIV self-test kits have the potential to increase testing rates among college students? Does offering STD kits improve testing rates among college students? How does stigma and lack of access impact STD rates among college students?	Sexually active college students considered in review	N/A	Scoping review framework	N/A
Design	**Research Instrument**	**Instrument Psychometrics**	**Procedures for Data Collection**	**Study Results**	**Level of Evidence**	**Grade Recommendation**
Scoping review	Multiple research databases	Electronic database searching, grey literature searching, data screening, and data extraction	Criteria for inclusion (1) discuss STD kit and self-testing (2) be in English (3) talk about college students (4) include participants beyond the age of 18 (5) be set in the USA	Barriers to STD self-testing among college students which included access to care, education, and social support	Level 4	Grade B

Summary of studies (continued)

Author	**Year**	**Study Design**	**Sample Size**	**Key Findings**	**Strengths**	**Limitations**
USPSTF	2021	Systematic Review	N= Various	Screening reduces infection in at-risk populations such as adolescents, sexually active women	Rigorous review	Limited focus on men and older persons
Author	**Year**	**Study Design**	**Sample Size**	**Key Findings**	**Strengths**	**Limitations**
CDC	2021	Screening guidelines	Various populations	Routine screening for STDs such as chlamydia and gonorrhea. For sexually active individuals particularly at-risk groups such as adolescents and sexually active women <25.	Based on research and data	Does not cover all populations equally
Author	**Year**	**Study Design**	**Sample Size**	**Key Findings**	**Strengths**	**Limitations**
American Family Physician	2008	Review Article	N/a	Early detection of chlamydia through routine screening can reduce complications and transmission	Comprehensive guideline	Limited new data from 2008 onward

Appendix B

Key Stakeholders in Needs Assessment.

Team Member	**Role & Responsibility**
Dr. Matthew Fete PhD	Dean of Health of Health and Human Services- Sponsor of the project has direct ties to the University's president and provost can present ideas to the Universities leaders to implement change and get action plans activated. Essentially the connection between our organization and the university that is non-clinical in nature however has the power to make things move.
Dr. Zenebe Wondwossen	Collaborating Physician- A stakeholder in this project someone who can give input from a clinical aspect. Can give expertise on how he has handled issues like this in the past. Will assist in developing an action plan to present to the university. Will review and critique management plan.
Janet Teske DNP	Clinical Supervisor- From an organizational standpoint will be responsible for ensuring that we uphold our companies' values while performing this needs assessment. Will help identify performance issues if applicable.
Cassandra Brooks FNP-C	Certified Family Nurse Practitioner (Governors State University Wellness Center)- Since our wellness center is modeled after theirs will have Cassandra pull data from her charts to see if she is experiencing the same issues at GSU. Will brainstorm interventions to bridge the gap.
Megan Novak	Wellness Center Coordinator- Will be responsible for professional communications between our organization and the universities

	leadership. Will identify current protocols or educational opportunities already in place for students residing in the dormitory.
Dr. Christine Brown	CSU Director of counseling- Responsible for data collection on how many students come for behavioral health when referred from the wellness center. Will relay and interpret results.
Kerri Winston	Director of Housing and Resident Life- Responsible for issuing student resources educational materials, condoms provided by wellness center. Responsible for communications between the university and the students on campus.

Note. Team involved in needs assessment and support of project proposal.

Appendix C

Sexual Health Questionnaire Consent

The following is a Sexual Health Questionnaire. The intent of this questionnaire is to assess your sexual behaviors and sexual history. This survey is voluntary and confidential. The following questions are used for analytical purposes only. Your responses will not be connected to you, or your personal information. Your signature implies that you understand and agree to participate in this survey.

Notice of Privacy Practices and Rights and Responsibilities:

I understand that although my personal information will not be utilized for this survey, I have certain rights to privacy regarding my protected health information. These rights are given to me under the Health Insurance portability and Accountability act of 1996 (HIPPA).

I acknowledge that I agree to participate in this survey and Notice of Privacy Practices has been disclosed. I understand the Notice describes Advocate Aurora's privacy practices regarding the use and/or disclosure of patient health information. I acknowledge that I have received verbal and written notice regarding my rights and responsibilities as a participant of this survey.

By my signature below, I acknowledge that I have read understand and agree to be bound by the terms of this consent form. I have had the opportunity to ask questions, and that any questions have been satisfactorily answered.

Signature of Participant Date of Signature

Print Name of Participant

Appendix C

5 P's	SEXUAL HEALTH QUESTIONNAIRE
Partners	Are you currently having Sex of any kind-oral, vaginal, or anal—with anyone? Yes or No In the last couple months how many sex partners have you had? __ What is/are the gender(s) of your sex partner(s)? Male or female or both Do you or your partner(s) currently have other partners? Yes or No
Practices	**To help understand your STI risk, what kind of sex have you had recently?** Have you had vaginal sex? Yes or No If yes, do you use condoms? ☐Never ☐Sometimes ☐ Always Have you had anal sex? Yes or No If yes, do you use condoms? ☐ Never ☐ Sometimes ☐ Always Have you ever had oral sex? Yes or No
Protection from STI's	Do you and your partner discuss STI prevention? Yes or No Have you received HPV, Hepatitis A, or Hepatitis B vaccines? Yes or No Are you familiar with PrEp, a medication that can prevent HIV? Have you ever considered using it? Yes or No
Past History of STIs	Have you ever been tested for STIs? Yes or No Would you like to be tested? Yes or No Have you ever been diagnosed with a STI in the past? Yes or No If yes, when? _____ did you get treatment? Yes or No Have you had any symptoms that keep coming back? Yes or No Has your current partner test positive for any STIs? Yes or No Do you know your partners HIV status? Yes or No
Pregnancy Intention	Do you think you want to have children at some point? When do you think that might be? Are you and your partner using contraception or practicing any form of birth control? Do you need any information on birth control? Yes or No

Note. Adapted from the Centers for Disease Control and Prevention. A guide to taking sexual history (2023)

Appendix D

Pre-and Post Sexually Transmitted Disease (STD) Participation Consent Form

The following is a pre-and post sexually transmitted disease survey to assess the attitudes of sexually active college students towards STDs before and after sexual health education. This survey is voluntary and confidential. The following questions are used for analytical purposes only. Your responses will not be connected to you or your personal information. Proceeding with this survey implies that you understand and agree to participate in this survey.

Notice of Privacy Practices and Rights and Responsibilities:

I understand that although my personal information will not be utilized for this survey, I have certain rights to privacy regarding my protected health information. These rights are given to me under the Health Insurance portability and Accountability act of 1996 (HIPPA).

I acknowledge that I agree to participate in this survey and Notice of Privacy Practices has been disclosed. I understand the Notice describes Advocate Aurora's privacy practices regarding the use and/or disclosure of patient health information. I acknowledge that I have received verbal and written notice regarding my rights and responsibilities as a participant of this survey.

By my signature below, I acknowledge that I have read understand and agree to be bound by the terms of this consent form. I have had the opportunity to ask questions, and that any questions have been satisfactorily answered.

Signature of Participant **Date of Signature**

Print Name of Participant

Appendix D

Pre- STD educational Survey

This survey is confidential. The following questions are used for analytical purposes only. Your responses will only be connected to the demographical data you provide. Proceeding with this survey implies that you understand and agree to this disclaimer.

Are you Hispanic, or Latino Origin? Yes or No

What is your race?

a. Black/African American
b. Asian
c. American Indian or Alaskan Native
d. Other Pacific Islander
e. Caucasian

Age_____ Gender_____ Year in college_____

Please rate your level of agreement with each statement.

1. I always use condoms.
   a. Strongly Disagree
   b. Disagree
   c. Neither Disagree nor Agree
   d. Strongly Agree
   e. Agree
2. I have been tested for Sexually Transmitted diseases.
   a. Strongly Disagree
   b. Disagree
   c. Neither Disagree nor Agree
   d. Strongly Agree
   e. Agree
3. I encourage my partner to be tested.
   a. Strongly Disagree
   b. Disagree
   c. Neither Disagree nor Agree
   d. Strongly Agree
   e. Agree
4. All sexually Transmitted Diseases are preventable.
   a. Strongly Disagree
   b. Disagree
   c. Neither Disagree nor Agree

d. Strongly Agree
e. Agree

5. I am confident in my knowledge on Sexually Transmitted Infections

a. Strongly Disagree
b. Disagree
c. Neither Disagree nor Agree
d. Strongly Agree
e. Agree

Note. Pre-STD Educational Survey used for the STD- educational intervention.

Post STD educational Survey

This survey is confidential. The following questions are used for analytical purposes only. Your responses will only be connected to the demographical data you provide. Proceeding with this survey implies that you understand and agree to this disclaimer.

Please rate your level of agreement with each statement. After learning about Sexually Transmitted Diseases.

1. I always use condoms.

    a. Strongly Disagree
    b. Disagree
    c. Neither Disagree nor Agree
    d. Strongly Agree
    e. Agree

2. I have been tested for Sexually Transmitted diseases.

    a. Strongly Disagree
    b. Disagree
    c. Neither Disagree nor Agree
    d. Strongly Agree
    e. Agree

3. I encourage my partner to be tested.
    a. Strongly Disagree
    b. Disagree
    c. Neither Disagree nor Agree
    d. Strongly Agree
    e. Agree

4. All sexually Transmitted Diseases are preventable.
    a. Strongly Disagree
    b. Disagree
    c. Neither Disagree nor Agree
    d. Strongly Agree
    e. Agree

5. I am confident in my knowledge on Sexually Transmitted Infections
    a. Strongly Disagree
    b. Disagree
    c. Neither Disagree nor Agree
    d. Strongly Agree
    e. Agree

6. I will always use condoms with sexual intercourse

    a. Strongly Disagree
    b. Disagree
    c. Neither Disagree nor Agree
    d. Strongly Agree
    e. Agree

Note. Post-STD educational survey utilized for the STD-educational event.

Appendix E

Consent to Treatment and Privacy Form

AdvocateAuroraHealth

Consent to Treat, Payment and Notice of Privacy Practices
(Use for ambulatory, non-hospital services)

To Treat:
I, for myself, (or the patient named below) hereby consent to and authorize medical treatment which may include the performance of examinations, treatments and diagnostic procedures which the physicians(s) or other appropriate providers at Advocate Aurora Health have advised me of and deemed medically necessary. I understand that health care providers in training may, under the supervision of appropriate personnel, participate in my treatment.

Release of Information:
I hereby authorize Advocate Aurora Health to release any medical information deemed necessary to process insurance claims (including information relating to the treatment of drug abuse, alcohol abuse, or mental illness).

Authorization of Payment:
I further authorize payment of any health insurance benefits directly to Advocate Aurora Health for services rendered to me or my dependent. This authorization applies to any insurance benefit that was in effect at the time the services were provided.

Payment:
I hereby agree to pay in full any balance on my account in accordance with the Advocate Aurora Health Payment and Credit policies, which may include reasonable attorney's fees. The balance due includes provisions set by my insurance company such as copayments, deductibles, and "usual and customary" allowance. Advocate Aurora Health reserves the right to change fees and policies without notice.

Independent Provider Services:
I acknowledge and fully understand that only those physicians/providers who are clearly identified as Advocate Aurora Health employees are agents of Advocate Aurora Health. Non-employed physicians/ providers are independent providers who are permitted to use Advocate Aurora facilities to render medical care and treatment. These independent physicians/providers exercise their own medical judgment in treating me or otherwise providing professional services to me. I understand that I should ask my physician any questions I have about his/ her employment status. My decision to seek medical care is not based upon any understanding, representation, advertisement, media campaign, inference, presumption or reliance that the physicians/providers caring for me are employees or agents of Advocate Aurora Health.

Notice of Privacy Practices and Rights and Responsibilities:
I acknowledge that Advocate Aurora Health has provided me a copy of its Notice of Privacy Practices. I understand the Notice describes Advocate Aurora's privacy practices regarding the use and/or disclosure of patient health information. I acknowledge that I have received verbal and written notice regarding my rights and responsibilities as a patient.

By my signature below, I acknowledge that I have read understand and agree to be bound by the terms of this consent form, including the specific language related to independent physician services. I have had the opportunity to ask questions, and that any questions have been satisfactorily answered.

Date _____ Time _____ Patient Signature _____

Date _____ Time _____ Legal Representative _____ Relationship to Patient _____

Interpreter Assistance: If an interpreter assisted, please complete the following: Language: _____

Date _____ Time _____ Interpreter Name _____ ID# _____

* 5 3 0 0 0 0 7 *

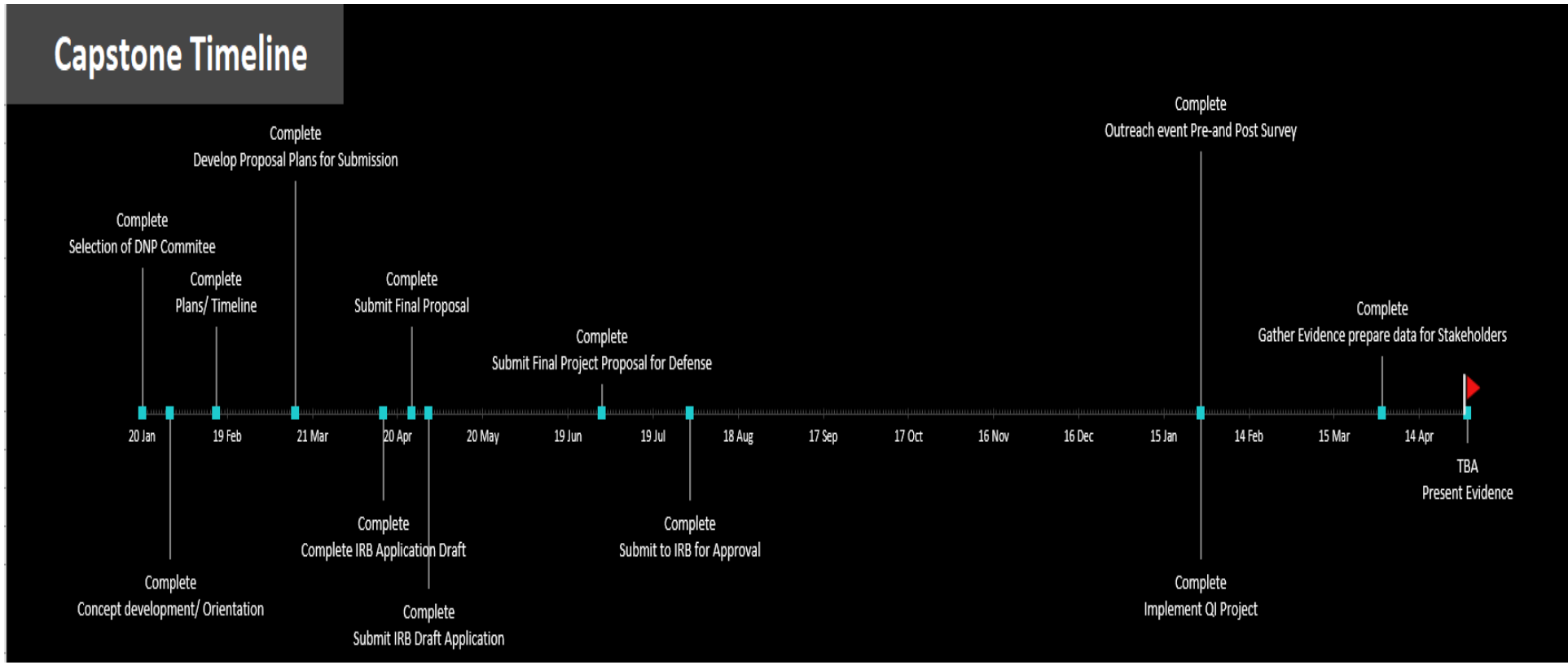
CONSENT TO TREAT, PAYMENT & NOTICE OF PRIVACY PRACTICES
(Consent – Ambulatory Treatment)

S300007 (09/22)

Note. Consent to Treatment and Privacy utilized for fast-track STD screening

Appendix F

Capstone Timeline



Note. This table demonstrates the capstone timeline with completed task.

Appendix F

Capstone Timeline Milestones

Project Milestones

Date	Milestone	Plan Duration
1/20/2023	Selection of DNP Commitee	Complete
1/30/2023	Concept development/ Orientation	Complete
2/15/2023	Plans/ Timeline	Complete
3/15/2023	Develop Proposal Plans for Submission	Complete
4/15/2023	Complete IRB Application Draft	Complete
4/25/2023	Submit Final Proposal	Complete
5/1/2023	Submit IRB Draft Application	Complete
7/1/2023	Submit Final Project Proposal for Defense	Complete
8/1/2023	Submit to IRB for Approval	Complete
1/28/2024	Implement QI Project	Complete
1/28/2024	Outreach event Pre-and Post Survey	Complete
4/1/2024	Gather Evidence prepare data for Stakeholders	Complete
5/1/2024	Present Evidence	TBA

Note. This figure demonstrates the capstone milestones and projected completion.

Appendix G

Acronyms and Abbreviations

APNs	Advanced Practice Nurses
CDC	Centers for Disease Control Disease and Prevention
CDPH	Chicago Department of Public Health
CINAHL	Cumulative Index to Nursing and Allied Health Literature
HBM	Health Belief Model
HIV	Human Immunodeficiency Virus
IBM	International Business Machines
IDPH	Illinois department of Public Health
JBI	Joanna Briggs Institute
PBC	Predominantly Black College
PHI	Protected Health Information
PICO	Population, Intervention, Control, Outcomes
STI	Sexually Transmitted Infection
STD	Sexually Transmitted Disease
SWOT	Strengths, Weakness, Opportunities, Threats
USPSTF	United States Preventive Services Task Force

Appendix H

Institutional Review Board (IRB) Approval Letters

Governors State UNIVERSITY

Institutional Review Board
Room G353
1 University Parkway
University Park, IL 60484
www.govst.edu/irb

Nov 21, 2023 4:42:38 PM CST

Shirley Spencer, Amanda Harris
Nursing

Re: Expedited Review - Initial submission - IRB-FY2024-34 A Pilot Program: Implementing a Fast-Track Sexually Transmitted Disease Screening Program to Decrease Chlamydia Cases on a College Campus

Dear Dr. Shirley Spencer:

The Governors State University Institutional Review Board has approved the Initial submission of your research protocol titled, "A Pilot Program: Implementing a Fast-Track Sexually Transmitted Disease Screening Program to Decrease Chlamydia Cases on a College Campus."

- All research related to this project **must** be conducted **exactly** as stated in the approved research protocol
- If you have completed the project, please complete a **Project Closure Submission**.
- If you would like to make any changes to research personnel or to the way that this research is conducted, you must completed a **Modification submission** and **wait for IRB approval** to enact any changes.
- Please be advised that you may **only** distribute the **GSU IRB approved** text, forms, documents, and materials to all participants. Your approved and stamped consent form can be found under the "Attachments" tab on the "Study Details" page.
- Please include the assigned IRB project number, PI name, and exact title of your project in any correspondence about this project.

Decision: Approved

Selected Category(ies): Expedited Category 5. Research involving materials (data, documents, records, or specimens) that have been collected, or will be collected solely for nonresearch purposes (such as medical treatment or diagnosis). (NOTE: Some research in this category may be exempt from the HHS regulations for the protection of human subjects. 45 CFR 46.101(b)(4). This listing refers only to research that is not exempt.)

7. Research on individual or group characteristics or behavior (including, but not limited to, research on perception, cognition, motivation, identity, language, communication, cultural beliefs or practices, and social behavior) or research employing survey, interview, oral history, focus group, program evaluation, human factors evaluation, or quality assurance methodologies. (NOTE: Some research in this category may be exempt from the HHS regulations for the protection of human subjects. 45 CFR 46.101(b)(2) and (b)(3). This listing refers only to research that is not exempt.)

Research Notes:

Sincerely,
Governors State University Institutional Review Board
irb@govst.edu

Appendix H

Institutional Review Board
FWA# 00002654
IRB # 00000807
9501 S. King Drive/ADM 304A
Chicago, IL 60628
773.995.5078
irb@csu.edu

December 16, 2023

Amanda Harris, APRN, FNP-BC
Nursing Department
Governors State University
1 University Parkway
University Park, IL. 60484

Re: Protocol #22-11-23, *A Pilot Program: Implementing a Fast-Track Sexually Transmitted Disease Screening Program to Decrease Chlamydia Cases on a College Campus*

Dear Amanda Harris,
The above referenced application has been reviewed and approved by the university's IRB under expedited review on December 16, 2023. As a low-risk project, this protocol is not subject to continuing review. However, please report any changes in the protocol or adverse events to the IRB immediately. Both forms are on the IRB website. You also must turn in a termination report when the project ends. You may also receive a request for an annual update on the project. Your research records may be audited at any time during or after the implementation of your project. Federal and University policies require that all research records be maintained for a period of three (3) years following the close of a research project.

Good luck with your project.
Sincerely,

Melany P. Puglisi, Ph.D.
Chair, CSU Institutional Review Board

Scholarship & Responsibility

csu.edu

CHICAGO STATE UNIVERSITY

Appendix I

COLLABORATIVE INSTITUTIONAL TRAINING INITIATIVE (CITI PROGRAM)
COMPLETION REPORT - PART 1 OF 2
COURSEWORK REQUIREMENTS*

* NOTE: Scores on this Requirements Report reflect quiz completions at the time all requirements for the course were met. See list below for details. See separate Transcript Report for more recent quiz scores, including those on optional (supplemental) course elements.

- **Name:** Amanda Harris (ID: 7742934)
- **Institution Affiliation:** Governors State University (ID: 867)
- **Institution Email:** aharris14@student.govst.edu
- **Phone:** 708-990-7018

- **Curriculum Group:** Human Research
- **Course Learner Group:** IRB Social/Behavioral Research
- **Stage:** Stage 1 - Basic Course

- **Record ID:** 53764380
- **Completion Date:** 31-Jan-2023
- **Expiration Date:** 31-Jan-2026
- **Minimum Passing:** 80
- **Reported Score*:** 93

REQUIRED AND ELECTIVE MODULES ONLY	DATE COMPLETED	SCORE
Belmont Report and Its Principles (ID: 1127)	19-Jan-2023	3/3 (100%)
Students in Research (ID: 1321)	19-Jan-2023	5/5 (100%)
History and Ethical Principles - SBE (ID: 490)	26-Jan-2023	5/5 (100%)
Defining Research with Human Subjects - SBE (ID: 491)	31-Jan-2023	5/5 (100%)
The Federal Regulations - SBE (ID: 502)	31-Jan-2023	4/5 (80%)
Assessing Risk - SBE (ID: 503)	31-Jan-2023	5/5 (100%)
Informed Consent - SBE (ID: 504)	31-Jan-2023	5/5 (100%)
Privacy and Confidentiality - SBE (ID: 505)	31-Jan-2023	5/5 (100%)
Research with Prisoners - SBE (ID: 506)	31-Jan-2023	5/5 (100%)
Research with Children - SBE (ID: 507)	31-Jan-2023	4/5 (80%)
Research in Public Elementary and Secondary Schools - SBE (ID: 508)	19-Jan-2023	4/5 (80%)
Internet-Based Research - SBE (ID: 510)	19-Jan-2023	4/5 (80%)
Governors State University Module (ID: 1638)	19-Jan-2023	No Quiz

For this Report to be valid, the learner identified above must have had a valid affiliation with the CITI Program subscribing institution identified above or have been a paid independent Learner.

Verify at: www.citiprogram.org/verify/?kbd37ef4c-11ac-4bc3-8fa4-d2366fe9c0bb-53764380

Collaborative Institutional Training Initiative (CITI Program)
101 NE 3rd Avenue
Suite 320
Fort Lauderdale, FL 33301 US

Email: support@citiprogram.org
Phone: 888-529-5929
Web: https://www.citiprogram.org

Appendix I

COLLABORATIVE INSTITUTIONAL TRAINING INITIATIVE (CITI PROGRAM)
COMPLETION REPORT - PART 2 OF 2
COURSEWORK TRANSCRIPT**

** NOTE: Scores on this Transcript Report reflect the most current quiz completions, including quizzes on optional (supplemental) elements of the course. See list below for details. See separate Requirements Report for the reported scores at the time all requirements for the course were met.

- **Name:** Amanda Harris (ID: 7742934)
- **Institution Affiliation:** Governors State University (ID: 867)
- **Institution Email:** aharris14@student.govst.edu
- **Phone:** 708-990-7018

- **Curriculum Group:** Human Research
- **Course Learner Group:** IRB Social/Behavioral Research
- **Stage:** Stage 1 - Basic Course

- **Record ID:** 53764380
- **Report Date:** 17-Oct-2023
- **Current Score**:** 93

REQUIRED, ELECTIVE, AND SUPPLEMENTAL MODULES	MOST RECENT	SCORE
History and Ethical Principles - SBE (ID: 490)	26-Jan-2023	5/5 (100%)
Defining Research with Human Subjects - SBE (ID: 491)	31-Jan-2023	5/5 (100%)
The Federal Regulations - SBE (ID: 502)	31-Jan-2023	4/5 (80%)
Belmont Report and Its Principles (ID: 1127)	19-Jan-2023	3/3 (100%)
Assessing Risk - SBE (ID: 503)	31-Jan-2023	5/5 (100%)
Informed Consent - SBE (ID: 504)	31-Jan-2023	5/5 (100%)
Privacy and Confidentiality - SBE (ID: 505)	31-Jan-2023	5/5 (100%)
Research with Prisoners - SBE (ID: 506)	31-Jan-2023	5/5 (100%)
Research with Children - SBE (ID: 507)	31-Jan-2023	4/5 (80%)
Research in Public Elementary and Secondary Schools - SBE (ID: 508)	19-Jan-2023	4/5 (80%)
Internet-Based Research - SBE (ID: 510)	19-Jan-2023	4/5 (80%)
Students in Research (ID: 1321)	19-Jan-2023	5/5 (100%)
Governors State University Module (ID: 1638)	19-Jan-2023	No Quiz

For this Report to be valid, the learner identified above must have had a valid affiliation with the CITI Program subscribing institution identified above or have been a paid independent Learner.

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